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UNIVERSITI TEKNOLOGI MALAYSIA



CAIRO
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JETSON AI BPLTV-KPM WORKSHOP JETSON, GIT, ANACONDA & PYTHON PROGRAMMING (PART 2)

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WHAT IS GitHub?



WHAT IS GitHub?

- Definition: GitHub is a web-based platform for version control and collaboration, built on top of Git.
- Purpose: It allows multiple people to work on a project simultaneously, track changes, and manage code repositories efficiently.
- Features: Hosting for Git repositories, collaboration tools, issue tracking, continuous integration and deployment (CI/CD) with GitHub Actions, and more.



BENEFITS OF USING GitHub

1. Building a Professional Portfolio

- **Showcase Your Work:** Use GitHub repositories to showcase projects, demonstrating your coding skills and project management abilities to potential employers.
- **Contributions:** Highlight contributions to open-source projects, showing collaboration and teamwork skills.

2. Learning from Open-Source Projects

- **Access to Code:** Explore and learn from the code of experienced developers by browsing popular open-source repositories.
- **Contributing:** Improve your skills by contributing to open-source projects, which also helps build your reputation in the developer community.

3. Collaboration and Teamwork

- **Working in Teams:** GitHub is widely used in professional settings for collaborative work. Learning to use it effectively prepares you for teamwork in real-world projects.
- **Feedback and Code Reviews:** Engage in code reviews and discussions, receiving feedback and improving your code quality.

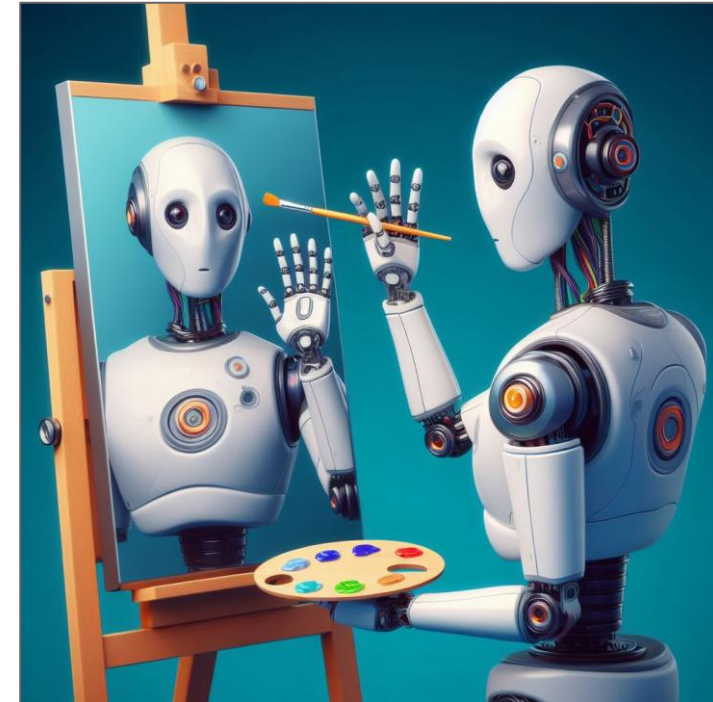
4. Project Management Skills

- **Issues and Project Boards:** Manage tasks, bugs, and project milestones using GitHub Issues and Project Boards, enhancing your project management skills.
- **Workflow Automation:** Use GitHub Actions to automate repetitive tasks, learning about CI/CD pipelines and improving productivity.

BENEFITS OF USING GitHub

5. Continuous Learning

- **Staying Updated:** Follow repositories and developers to stay updated on the latest technologies and trends in the software development industry.
- **GitHub Education:** Take advantage of resources and tools provided by GitHub Education for students.



UNDERSTANDING GIT & GitHub

Git: A Version Control System

- Git is a distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Key Features

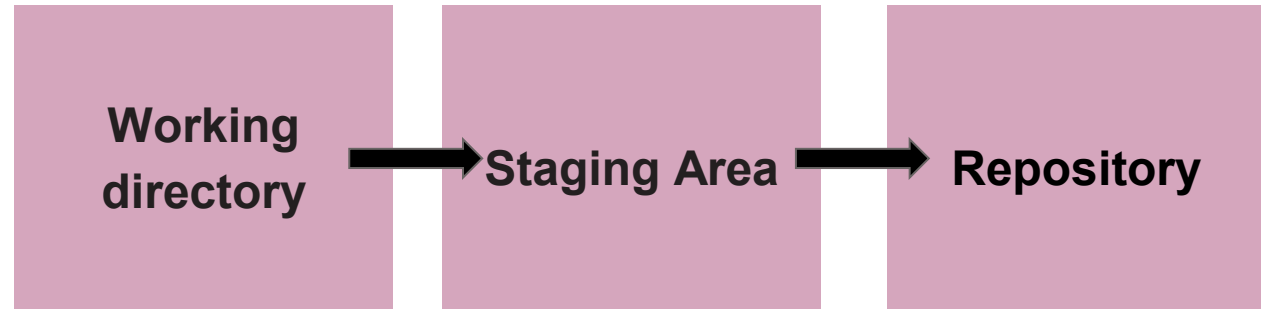
Distributed: Every developer has a full copy of the project history.

Branching and Merging: Support for nonlinear development through branches.

Performance: Optimized for high performance.

Security: Ensures the integrity of the code and history.

Workflow



Working Directory: The files in your project.

Staging Area: A place to store changes before committing them.

Repository: The history of all commits and branches.

UNDERSTANDING GIT & GitHub



GitHub: A Cloud-Based Hosting Service for Git Repositories

- GitHub is a platform for hosting and collaborating on Git repositories.



GitHub vs. Other Git Hosting Services:

- GitHub: Known for its user-friendly interface, community engagement, and GitHub Actions for Continuous Integration (CI)/ Continuous Deployment/Delivery(CD).
- Alternatives: GitLab, Bitbucket, and others, each with unique features and strengths.

Key features

1. Collaboration: Tools for code review, issue tracking, and project management.
2. Integration: Supports integration with CI/CD tools and other development tools.
3. Community: Millions of developers and repositories available for collaboration and learning.

RELATIONSHIP OF GIT & GitHub



- **Git:**

Tool: Command-line tool installed locally on your computer.

Function: Version control system to track changes in your project.

- **GitHub:**

Service: Online platform that hosts Git repositories.

Function: Facilitates collaboration, code sharing, and project management.

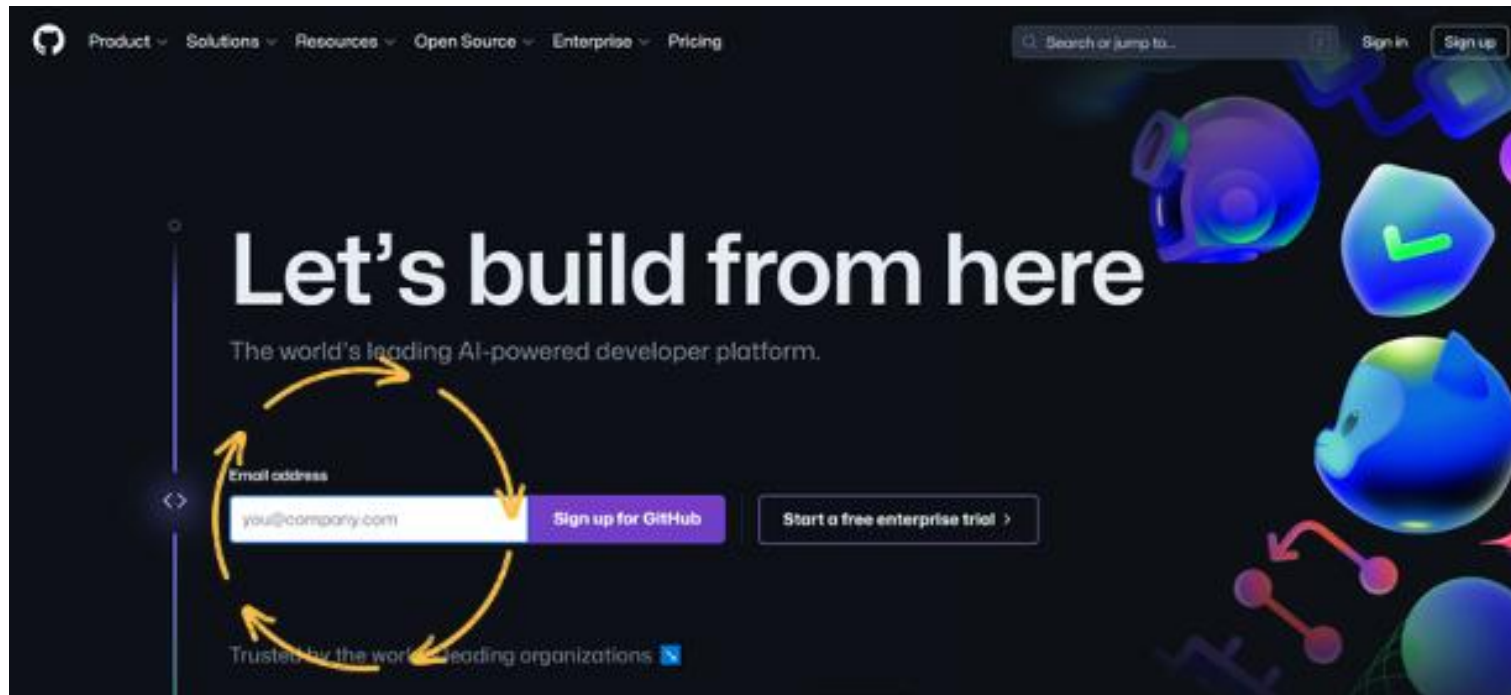
- **Relationship:**

Workflow: Developers use Git to track changes locally and push/pull changes to/from GitHub for collaboration.

SETTING UP GitHub

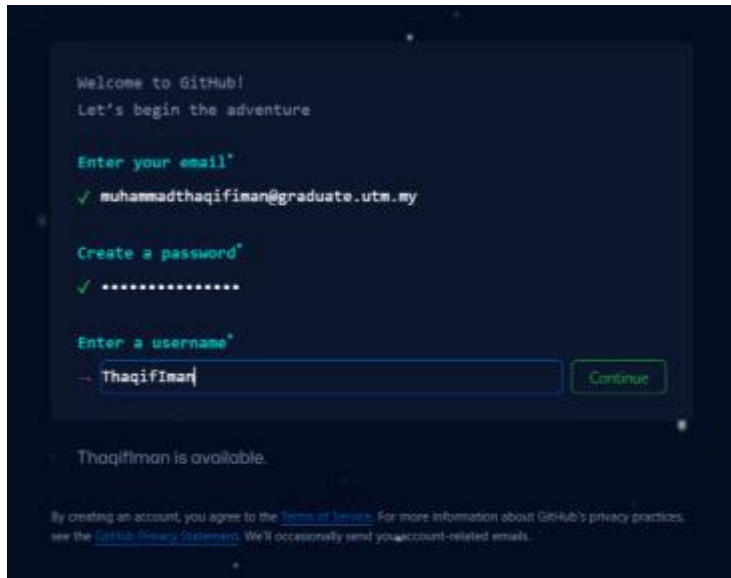
Creating a GitHub Account:

Sign Up: Go to GitHub's website and enter your username, email, and password. Click "SIGN UP". Follow the prompts to complete the registration.



SETTING UP GitHub

Email Verification: Verify your email address by clicking the link sent to your inbox.



Welcome to GitHub!
Let's begin the adventure

Enter your email*

✓ muhammadthaqifiman@graduate.utm.my

Create a password*

✓

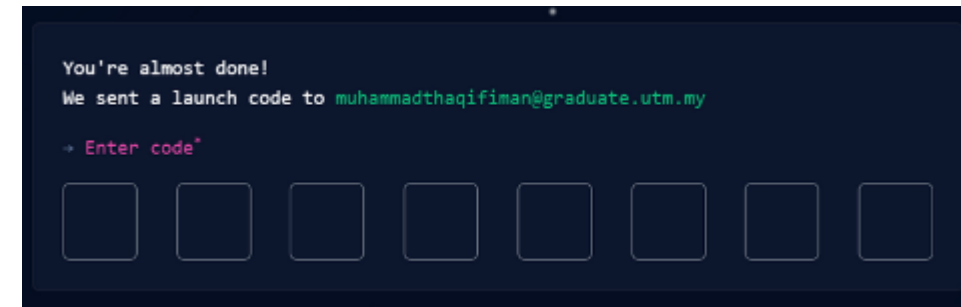
Enter a username*

ThaqifIman

Continue

ThaqifIman is available.

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.



You're almost done!

We sent a launch code to muhammadthaqifiman@graduate.utm.my

→ Enter code*

□ □ □ □ □ □ □ □

1. Creating a New Repository

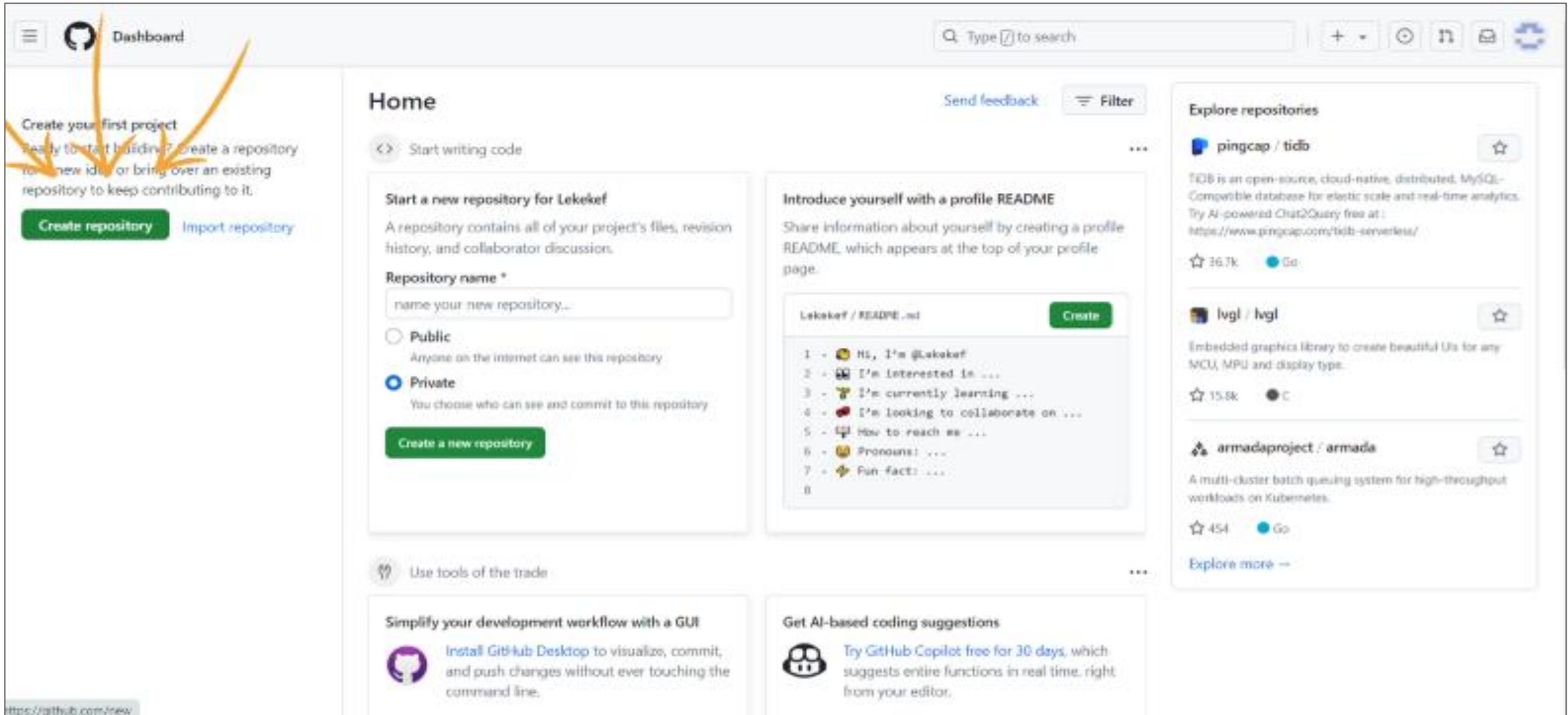
Steps:

- Navigate to Repositories: Click on the "Create Repositories" tab in your GitHub page.

Fill in Details:

- Repository Name: Choose a unique name for your repository.
- Description: Optionally, add a description of your repository.
- Public/Private: Choose whether the repository will be public or private.
- Initialize with a README: Check this box to add a README file.
- Create Repository: Click the "Create repository" button.

CREATING AND MANAGING REPOSITORIES



Dashboard

Create your first project
Ready to start building? Create a repository for a new idea or bring over an existing repository to keep contributing to it.

[Create repository](#) [Import repository](#)

Home

[Send feedback](#) [Filter](#)

[Start writing code](#)

Start a new repository for Lekekef

A repository contains all of your project's files, revision history, and collaborator discussion.

Repository name *

☐ Public
Anyone on the internet can see this repository

☒ Private
You choose who can see and commit to this repository

[Create a new repository](#)

Introduce yourself with a profile README

Share information about yourself by creating a profile README, which appears at the top of your profile page.

Lekekef / README.md [Create](#)

```
1 - 🐼 Hi, I'm @Lekekef
2 - 🐼 I'm interested in ...
3 - 🐼 I'm currently learning ...
4 - 🐼 I'm looking to collaborate on ...
5 - 🐼 How to reach me ...
6 - 🐼 Pronouns: ...
7 - 🐼 Fun fact: ...
8
```

Explore repositories

[pingcap / tidb](#) [Star](#)

TiDB is an open-source, cloud-native, distributed, MySQL-Compatible database for elastic scale and real-time analytics. Try AI-powered Chat2Query free at: <https://www.pingcap.com/tidb-serverless/>

☆ 36.7k [Go](#)

[lvgl / lvgl](#) [Star](#)

Embedded graphics library to create beautiful UIs for any MCU, MPU and display type.

☆ 15.8k [Go](#)

[armadaproject / armada](#) [Star](#)

A multi-cluster batch queuing system for high-throughput workloads on Kubernetes.

☆ 454 [Go](#)

[Explore more](#)

Simplify your development workflow with a GUI

[Install GitHub Desktop](#) to visualize, commit, and push changes without ever touching the command line.

Get AI-based coding suggestions

Try [GitHub Copilot](#) free for 30 days, which suggests entire functions in real time, right from your editor.

<https://github.com/new>

CREATING AND MANAGING REPOSITORIES



Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk ().*

Owner *

Repository name *

Lekekef

/ testinggg

testinggg is available.

Great repository names are short and memorable. Need inspiration? How about [symmetrical-octo-succotash](#)?

Description (optional)

This repo is for testing purpose onlyS

☒ Public

Anyone on the internet can see this repository. You choose who can commit.

☐ Private

You choose who can see and commit to this repository.

Initialize this repository with:

☒ Add a README file

This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

License: None

A license tells others what they can and can't do with your code. [Learn more about licenses.](#)

This will set `main` as the default branch. Change the default name in your [settings](#).

ⓘ

You are creating a public repository in your personal account.

Create repository

CREATING AND MANAGING REPOSITORIES



A screenshot of a GitHub repository page for a user named 'Lekekef' with the repository name 'testinggg'. The page is public. At the top, there's a navigation bar with links to Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. Below this, the repository name 'testinggg' is shown with a 'Public' badge. To the right of the name are buttons for Pin, Unwatch (1), Fork (0), Star (0), and a dropdown arrow. Below the repository name, there's a section for branches and tags, showing 'main' as the selected branch, '1 Branch', and '0 Tags'. A search bar 'Go to file' and buttons 'Add file' and 'Code' are present. The main content area shows a commit history table with one entry: 'Initial commit' by 'Lekekef' at '316d0de' 'now'. Below the table is a 'README' section with the title 'testinggg' and the text 'This repo is for testing purpose only\$'. On the right side, there's an 'About' section with the text 'This repo is for testing purpose only\$'. Below 'About' are links for 'Readme', 'Activity', '0 stars', '1 watching', and '0 forks'. Further down are sections for 'Releases' (No releases published, Create a new release) and 'Packages' (No packages published, Publish your first package). At the bottom, there's a footer with the GitHub logo, copyright notice '© 2024 GitHub, Inc.', and links to Terms, Privacy, Security, Status, Docs, Contact, Manage cookies, and Do not share my personal information.

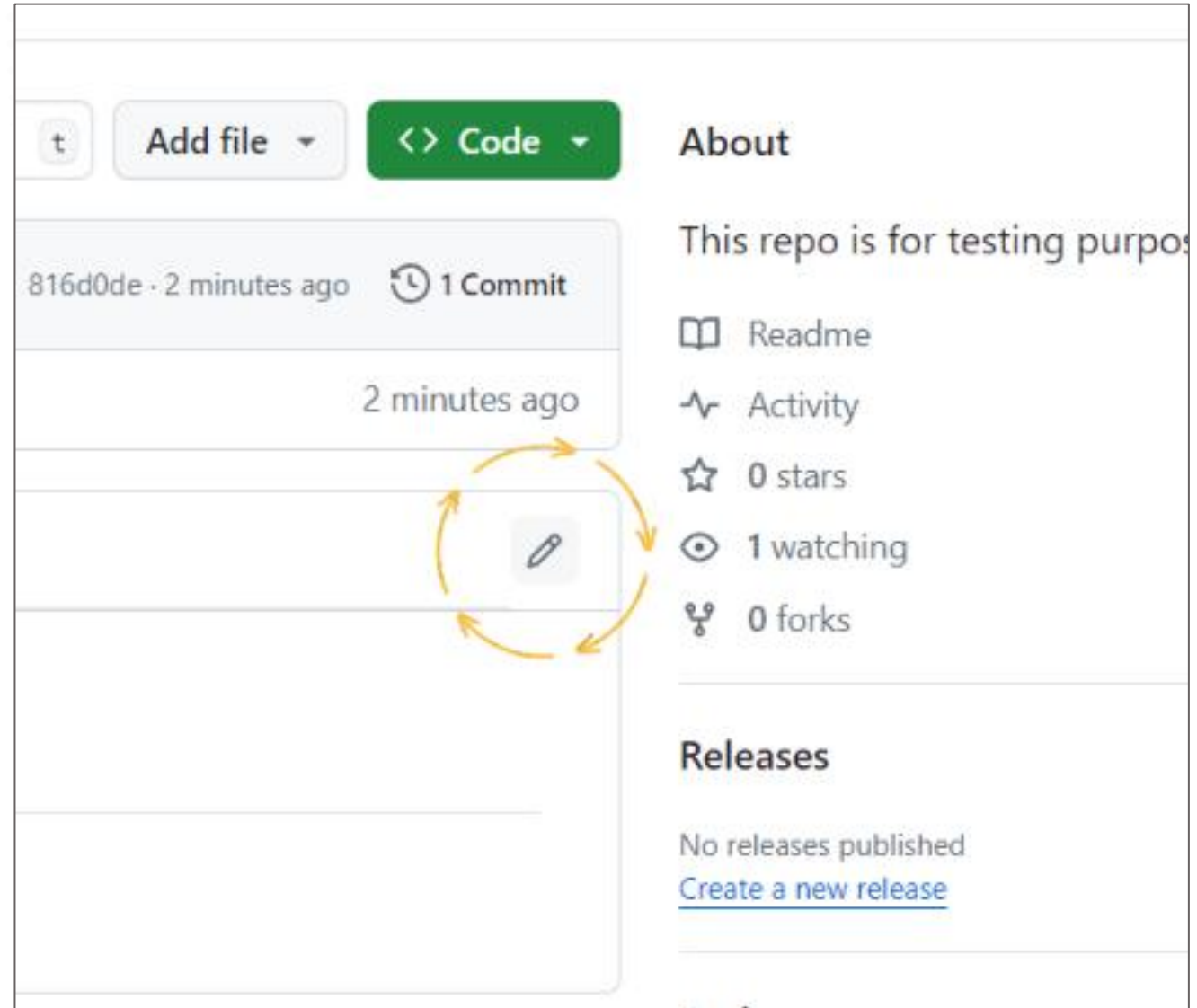
2. README File

Purpose: A README file provides information about your project, such as what it does, how to use it, and how to contribute.

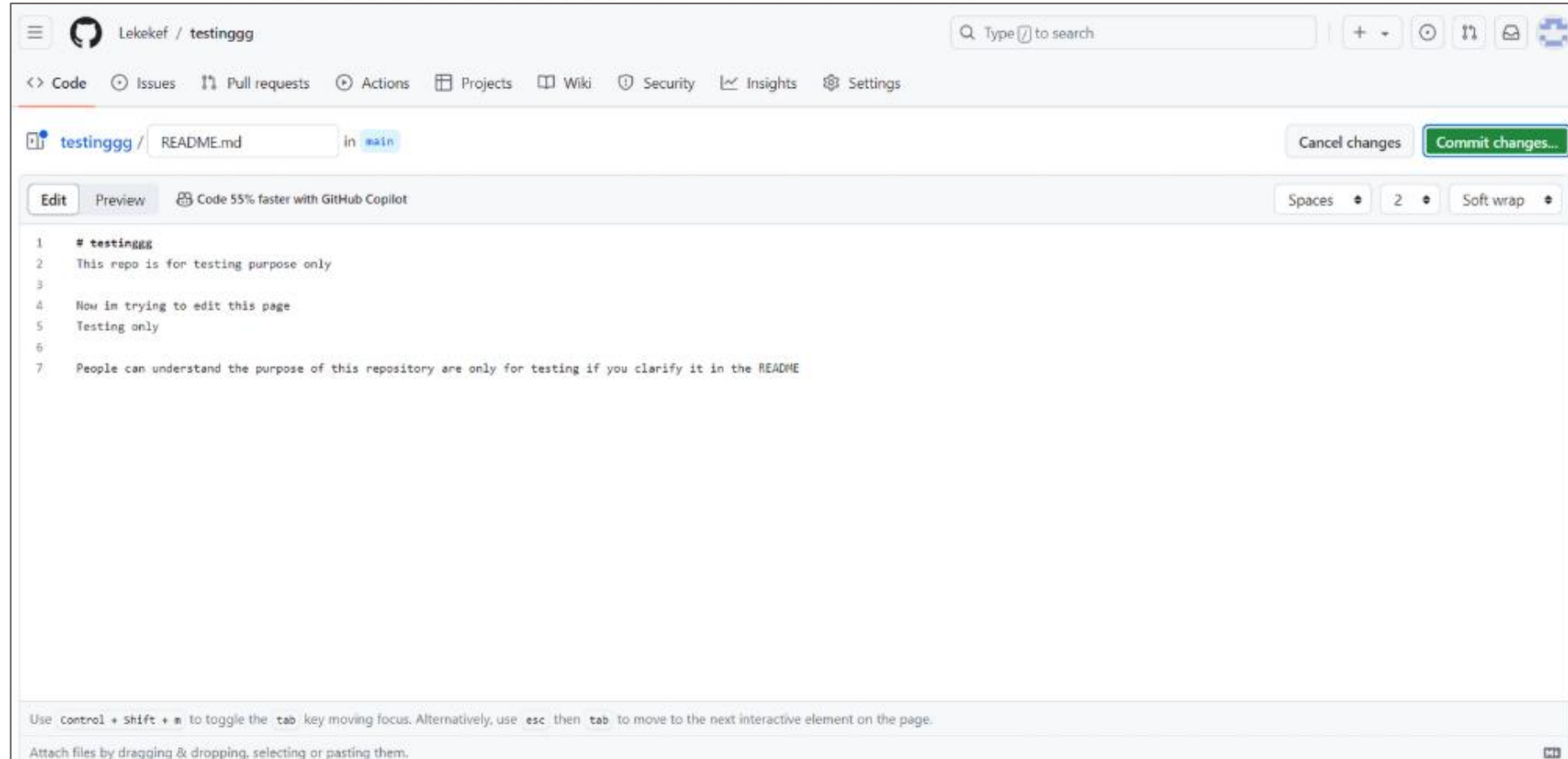
Steps:

1. Navigate to Repository: Go to your newly created repository.
2. Add README: If you didn't initialize with a README, click "Add a README" on the repository home page.
3. Edit README: Use the GitHub editor to write and format your README in Markdown.
4. Commit Changes: Click "Commit changes" to save your changes.

EDIT BUTTON HERE
click this to edit your
README

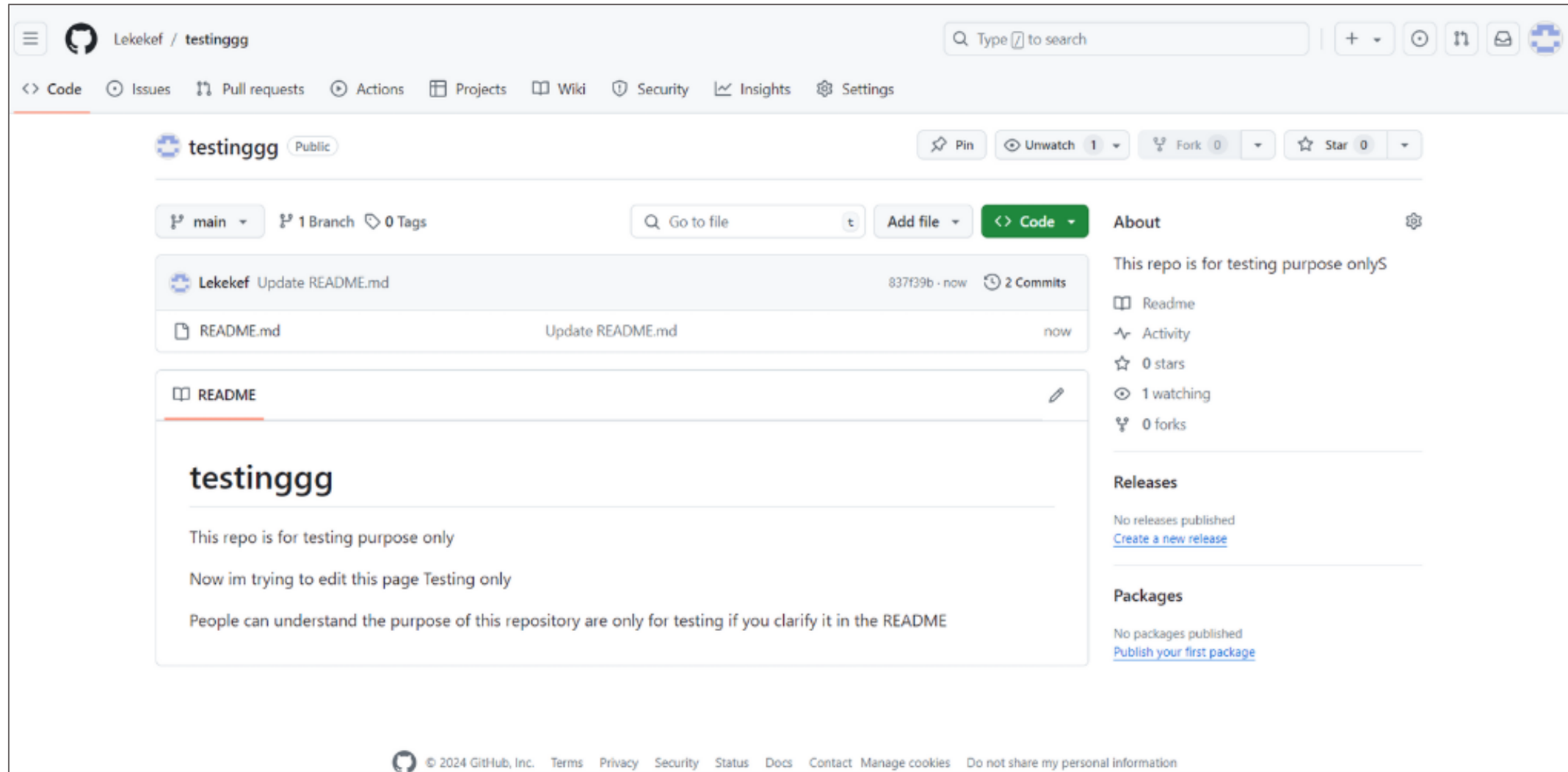


CREATING AND MANAGING REPOSITORIES



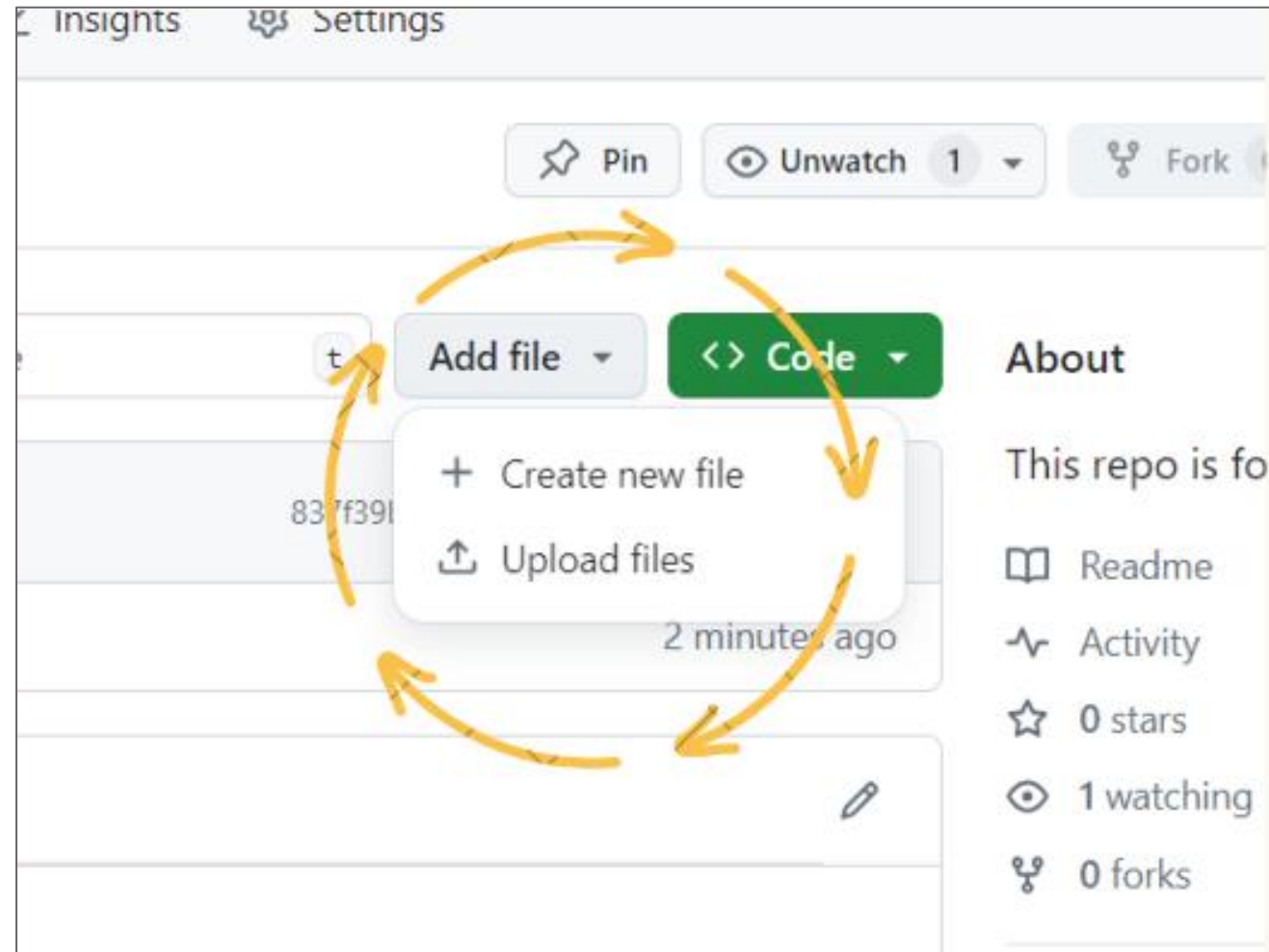
**You may try and edit your README to give clear descriptions of your projects.
Once done, click “Commit Changes” to save your edits.**

CREATING AND MANAGING REPOSITORIES



You can view the changes on the main page

Click here to try and
UPLOAD FILES



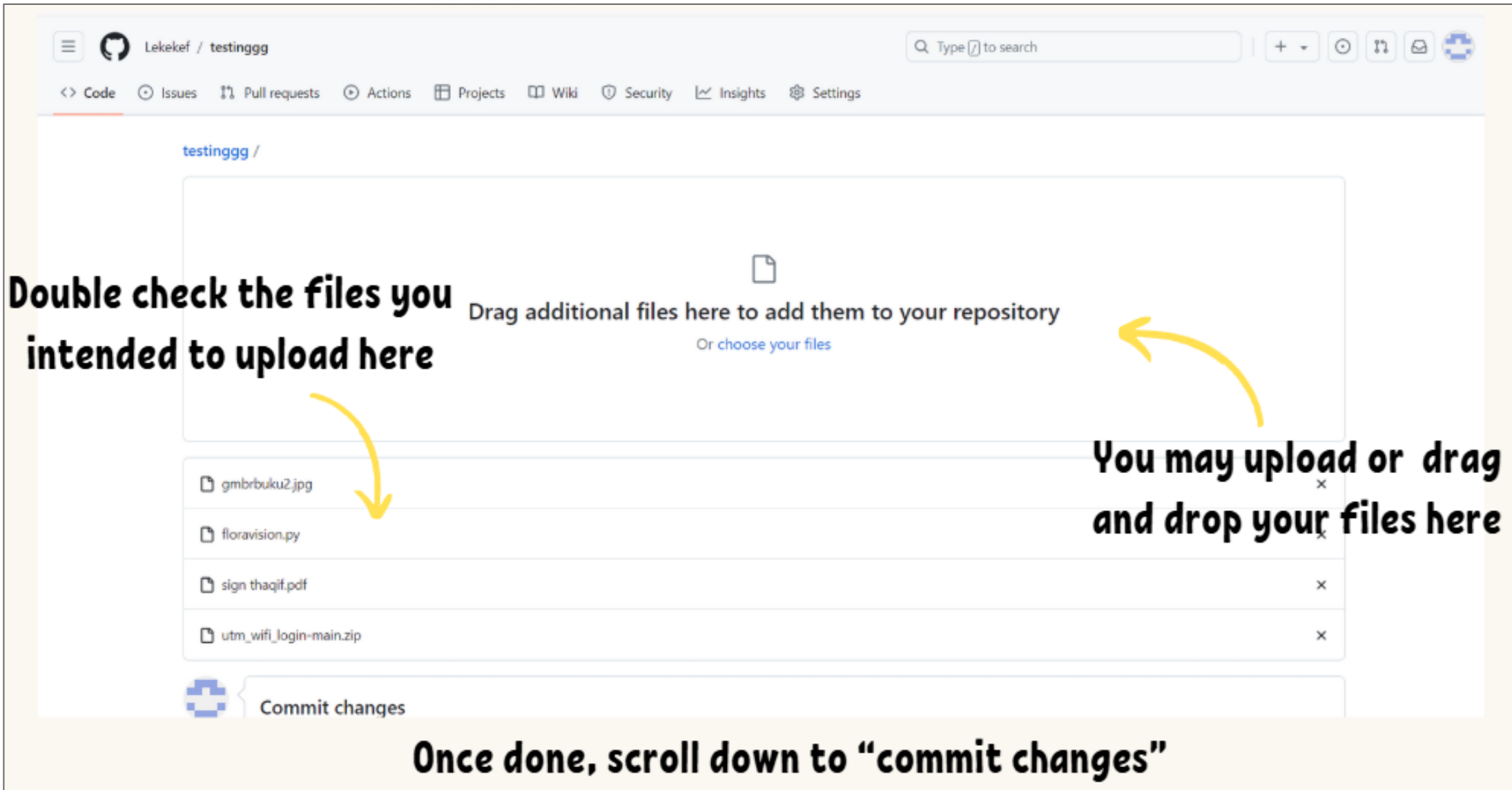
CREATING AND MANAGING REPOSITORIES

A screenshot of a GitHub repository editor interface. The repository is named "testingggg" and the file being edited is "requirements.txt". The file content is as follows:

```
1  numpy
2  os
3  matplotlib
4  torchvision
5  cv2
```

The interface includes a top navigation bar with links to Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. A search bar is located in the top right. The editor has tabs for "Edit" and "Preview". On the right side, there are buttons for "Cancel changes" and "Commit changes...". Three yellow arrows with text annotations are overlaid on the image: 1. An arrow points from the text "1.Name your file" to the file name "requirements.txt" in the breadcrumb. 2. An arrow points from the text "2.Edit text into the file" to the code content area. 3. An arrow points from the text "3. 'Commit changes' to save your file" to the "Commit changes..." button. A footer note at the bottom of the editor states: "Use `Control + Shift + =` to toggle the `tab` key moving focus. Alternatively, use `esc` then `tab` to move to the next interactive element on the page."

CREATING AND MANAGING REPOSITORIES



The screenshot shows the GitHub web interface for a repository named 'testinggg' by user 'Lekekef'. The top navigation bar includes links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. A search bar is located on the right. The main content area displays a large box for uploading files with the text 'Drag additional files here to add them to your repository' and a link 'Or choose your files'. Below this, a list of files is shown: 'gmbrbuku2.jpg', 'floravision.py', 'sign thaqif.pdf', and 'utm_wifi_login-main.zip'. At the bottom, there is a 'Commit changes' button. Annotations with yellow arrows point to the upload area and the file list.

Double check the files you intended to upload here

Drag additional files here to add them to your repository
Or [choose your files](#)

You may upload or drag and drop your files here

gmbrbuku2.jpg
floravision.py
sign thaqif.pdf
utm_wifi_login-main.zip

 Commit changes

Once done, scroll down to “commit changes”

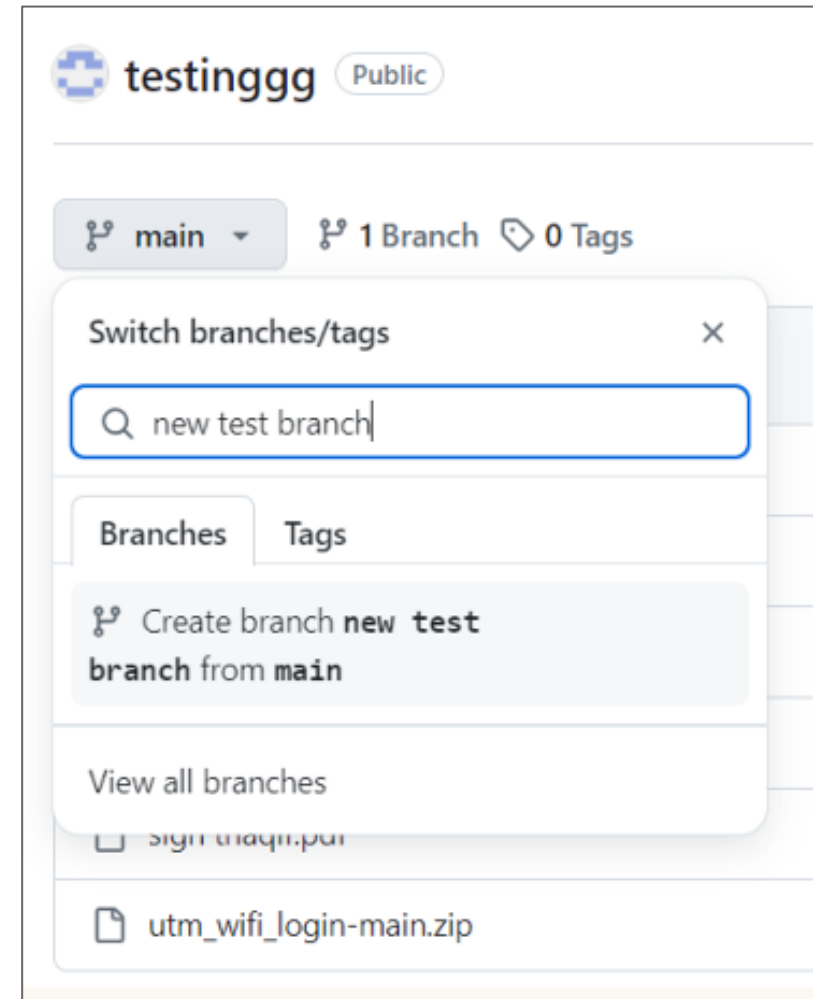
UNDERSTANDING BRANCHES

1. Creating Branches

Purpose: Branches allow you to develop features, fix bugs, or safely experiment with new ideas in a contained area.

Steps:

- Navigate to Repository: Go to your repository on GitHub.
- Branch Menu: Click the branch dropdown menu (usually shows "main" or "master").
- Create Branch:
Type a new branch name into the text box.
Press "Enter" to create the new branch.



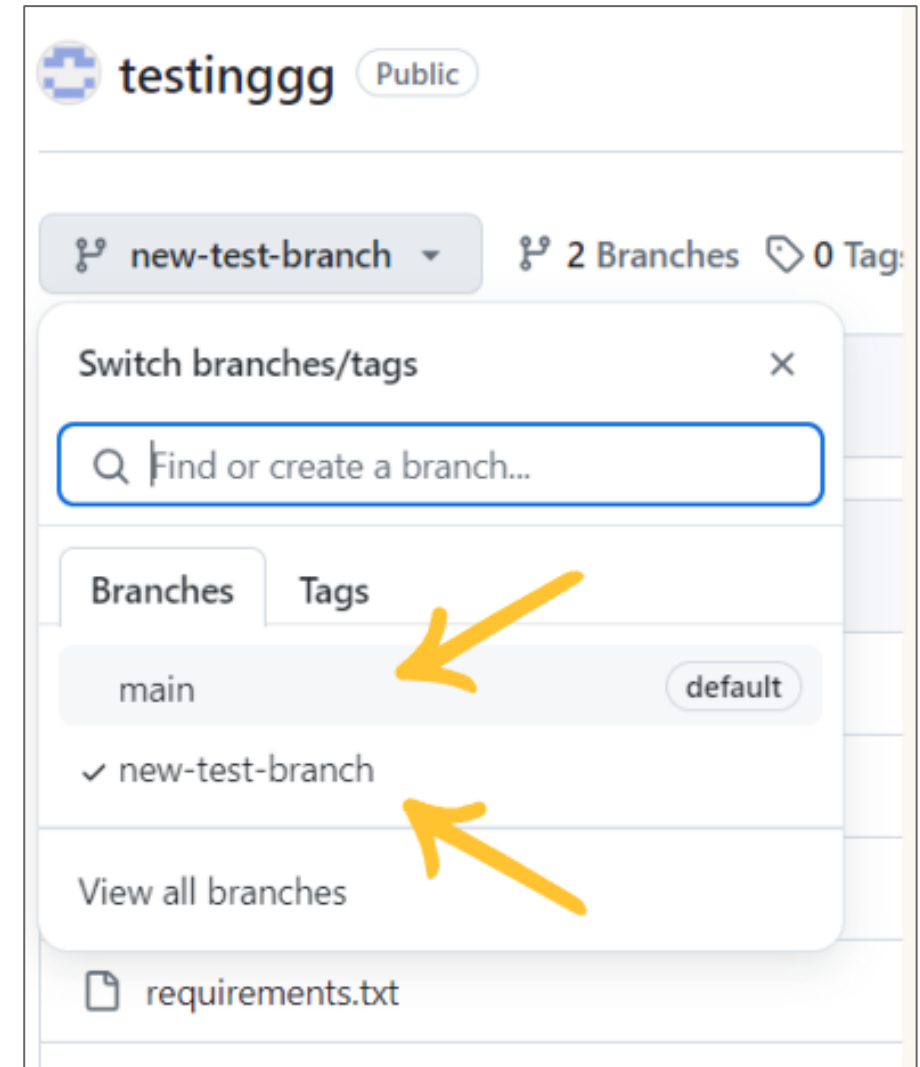
UNDERSTANDING BRANCHES

2. Switching Between Branches

Purpose: Switch between branches to view or work on different versions of your project.

Steps:

- Branch Menu: Click the branch dropdown menu.
- Select Branch: Click on the branch you want to switch to from the list.



3. Comparing Branches

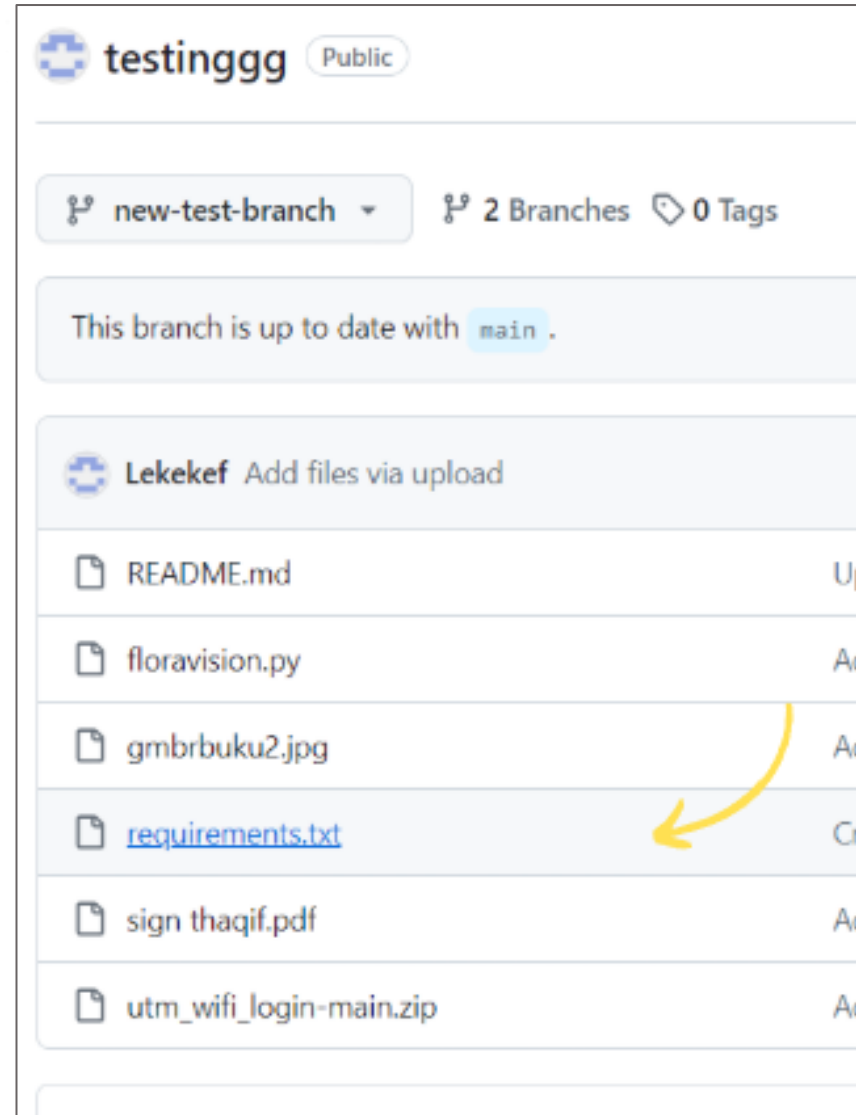
Purpose: Review the differences between branches to understand what changes have been made.

Steps:

- Navigate to Repository: Go to your repository on GitHub.
- Compare Button: Click the "Compare" button or navigate to the "Pull requests" tab and click "New pull request".
- Select Branches: Choose the base branch and compare branch.
- View Changes: GitHub will show a diff of the changes between the two branches.

UNDERSTANDING BRANCHES

For example here, lets try and edit the content of this file



UNDERSTANDING BRANCHES

testinggg / requirements.txt

Lekekef Create requirements.txt aa90f2a · 28 minutes ago History

Code Blame 5 lines (5 loc) · 36 Bytes Code 55% faster with GitHub Copilot Raw Copy Edit this file

```
1 numpy
2 os
3 matplotlib
4 torchvision
5 cv2
```

testinggg / requirements.txt in new-test-branch Cancel changes Commit changes...

Edit Preview Code 55% faster with GitHub Copilot Spaces 2 No wrap

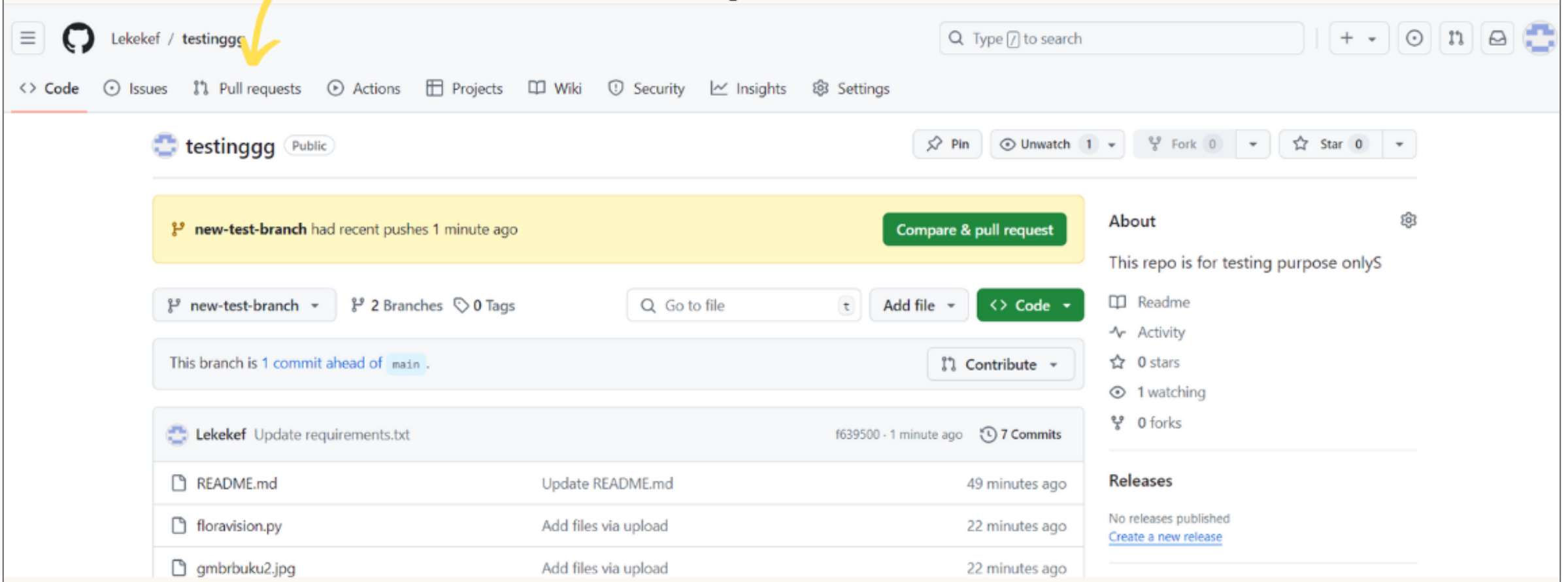
Type in the content/text/prompt you want to add

```
1 numpy
2 os
3 matplotlib
4 torchvision
5 cv2
6 pandas
7 seaborn
8 scipy
9 alumentations|
```

Click "Commit changes" to save your edits

UNDERSTANDING BRANCHES

To compare the edited file back with the original file, click on the “Pull Requests” tab



The screenshot shows the GitHub interface for the repository 'Lekekef / testinggg'. The 'Pull requests' tab is highlighted with a yellow arrow. The repository is public and has 2 branches and 0 tags. The current branch is 'new-test-branch', which is 1 commit ahead of 'main'. A yellow banner indicates that 'new-test-branch' had recent pushes 1 minute ago, with a 'Compare & pull request' button. The file list shows 'requirements.txt' updated 1 minute ago (commit f639500) and 'README.md', 'floravision.py', and 'gmbrbuku2.jpg' added via upload 22 minutes ago. The right sidebar shows repository statistics: 0 stars, 1 watching, and 0 forks.

Lekekef / testinggg

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

testinggg Public

new-test-branch had recent pushes 1 minute ago [Compare & pull request](#)

new-test-branch 2 Branches 0 Tags

This branch is 1 commit ahead of main.

Lekekef Update requirements.txt f639500 · 1 minute ago 7 Commits

README.md	Update README.md	49 minutes ago
floravision.py	Add files via upload	22 minutes ago
gmbrbuku2.jpg	Add files via upload	22 minutes ago

About

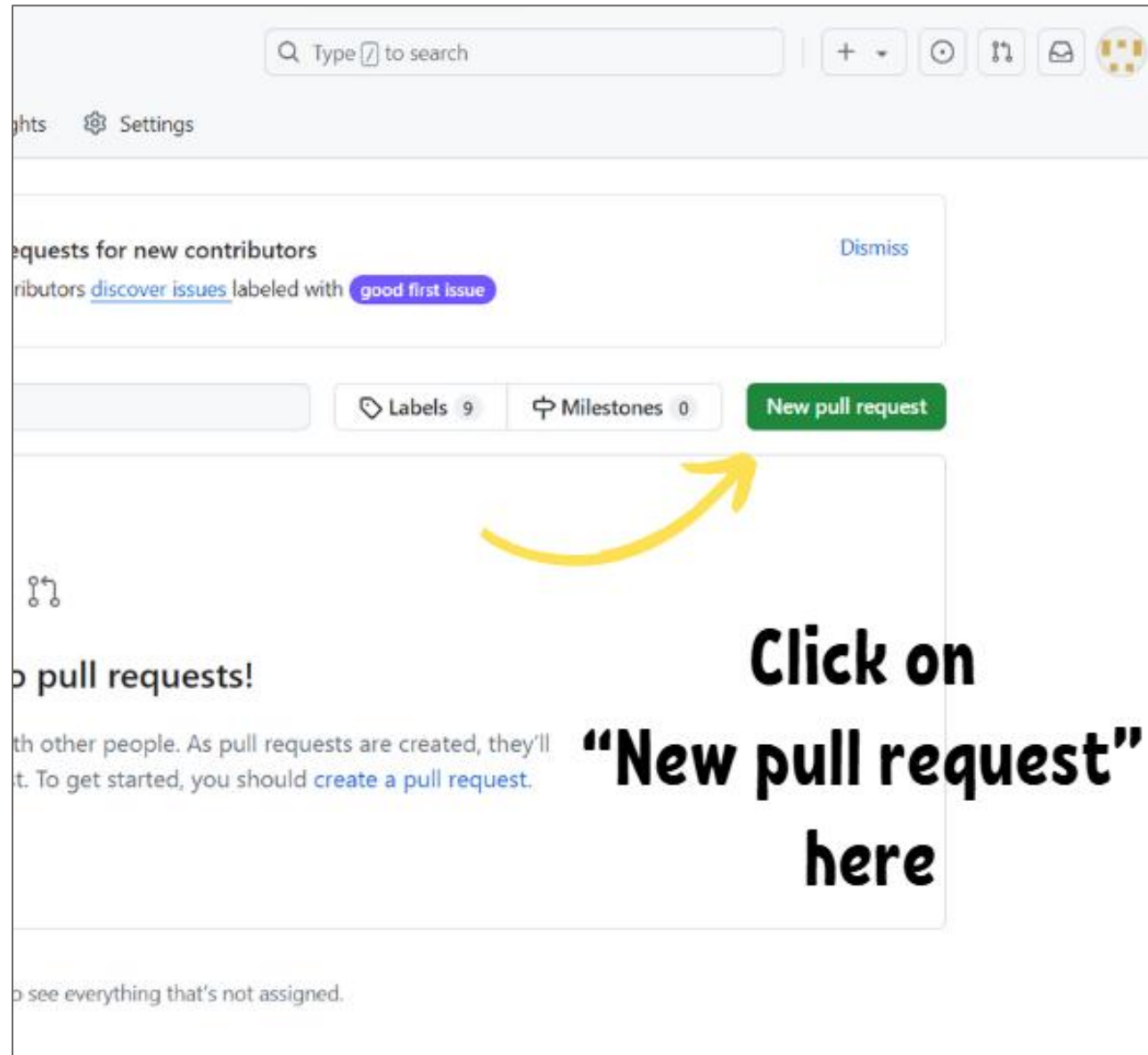
This repo is for testing purpose only

Readme Activity 0 stars 1 watching 0 forks

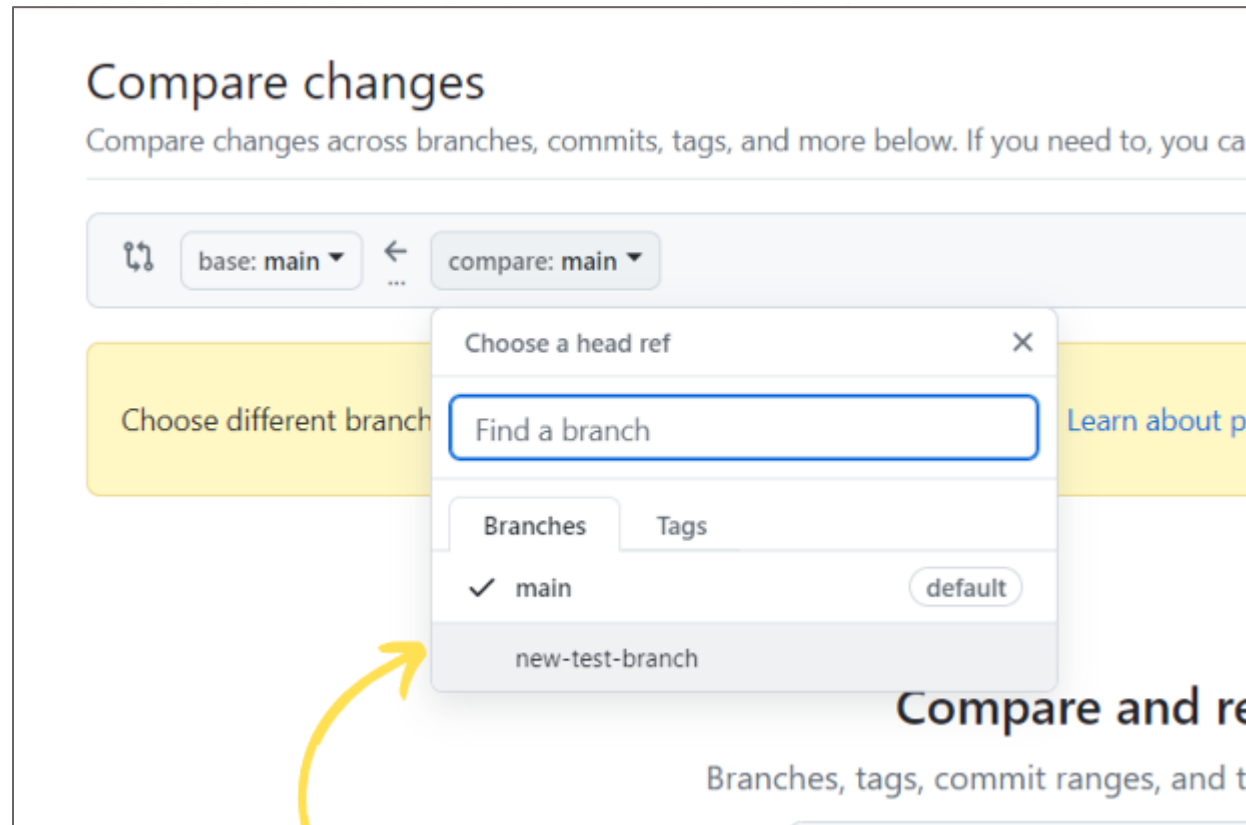
Releases

No releases published [Create a new release](#)

UNDERSTANDING BRANCHES



UNDERSTANDING BRANCHES



Next, we are going to compare the ‘edited file’ in the ‘new test branch’ with the ‘original file’ in the ‘main branch’. Choose the ‘new test branch’ here

UNDERSTANDING BRANCHES: EXAMPLE

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base: main ← compare: new-test-branch ✓ Able to merge. These branches can be automatically merged.

🔗 Update requirements.txt #1
No description available [View pull request](#)

1 commit 1 file changed 1 contributor

Commits on Aug 8, 2024

Update requirements.txt
Lekekef committed 1 hour ago Verified f639500 <>

Showing 1 changed file with 4 additions and 0 deletions. Split Unified

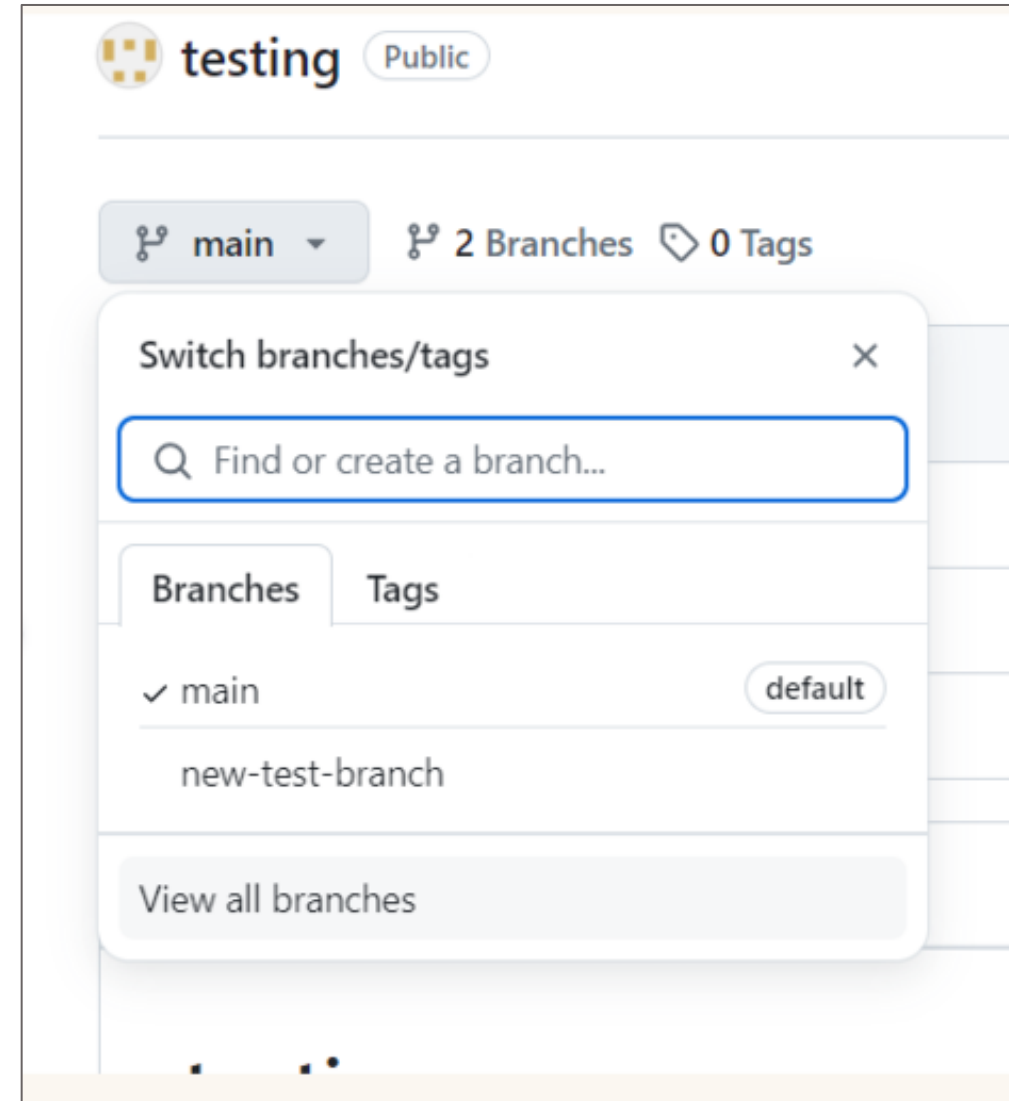
requirements.txt

↑	@@ -3,3 +3,7 @@ os
3	3 matplotlib
4	4 torchvision
5	5 cv2
	6 + pandas
	7 + seaborn
	8 + scipy
	9 + albumentations

Here, we may view the new input done in the 'new test branch' unified with the original one in the 'main branch'

4. Deleting Branches

- **Purpose:** Clean up unnecessary branches to keep repository organized
- **Step:**
 - Navigate to Repository: Go to your repository on GitHub.
 - Branches Tab: Click the "Branches" tab.
 - Delete Branch: Click the trash can icon next to the branch you want to delete. Note: You cannot delete the branch you are currently on.



UNDERSTANDING BRANCHES: DELETE BRANCH



Overview

Yours

Active

Stale

All

Q

Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
main	1 hour ago			Default	...

Your main branch isn't protected

Protect this branch from force pushing or deletion, or require status checks before merging. [Learn more](#)

Dismiss

Protect this branch

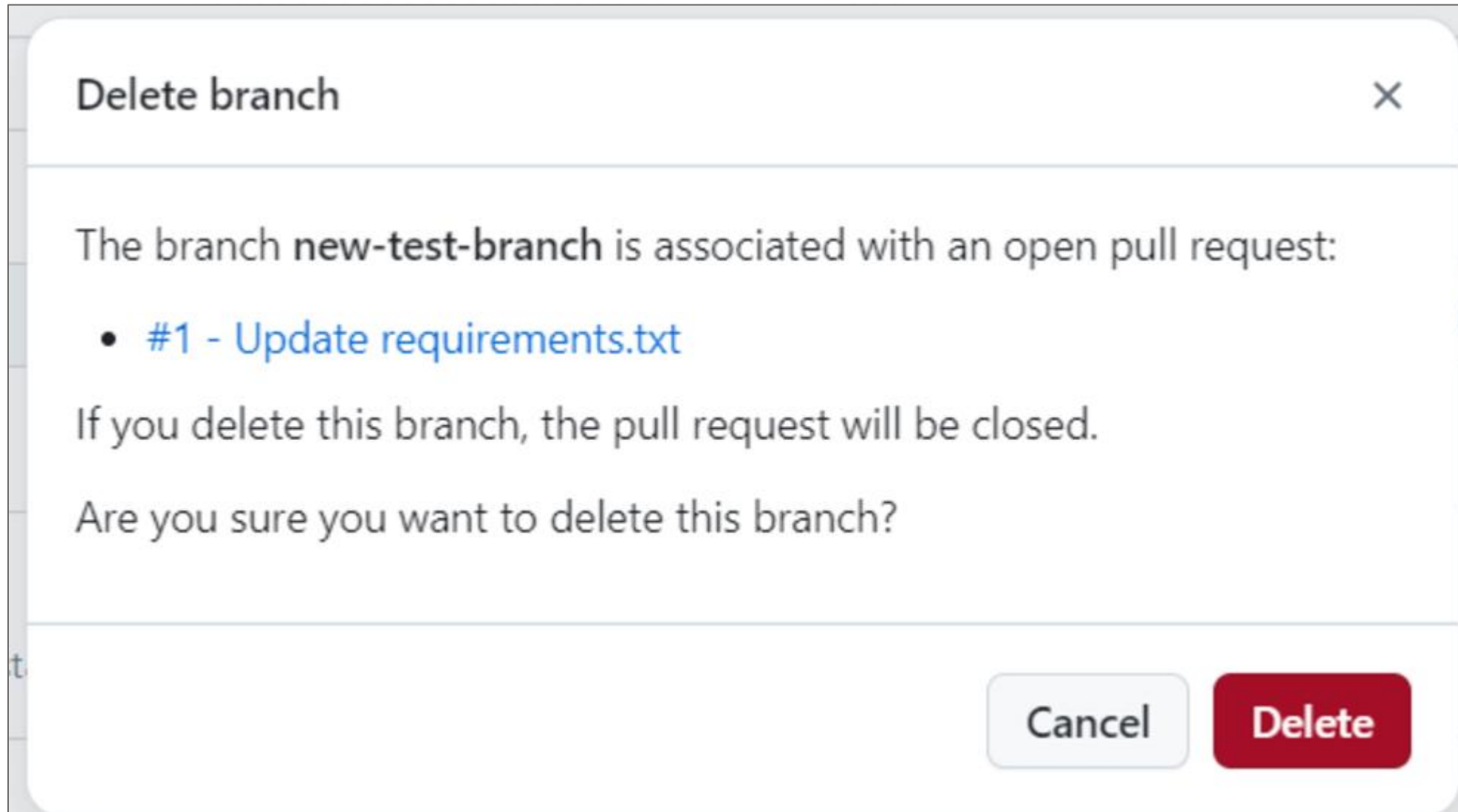
Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
new-test-branch	1 hour ago		0	1	#1 ...

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
new-test-branch	1 hour ago		0	1	#1 ...

Click on the 'trash can' icon to delete your branch



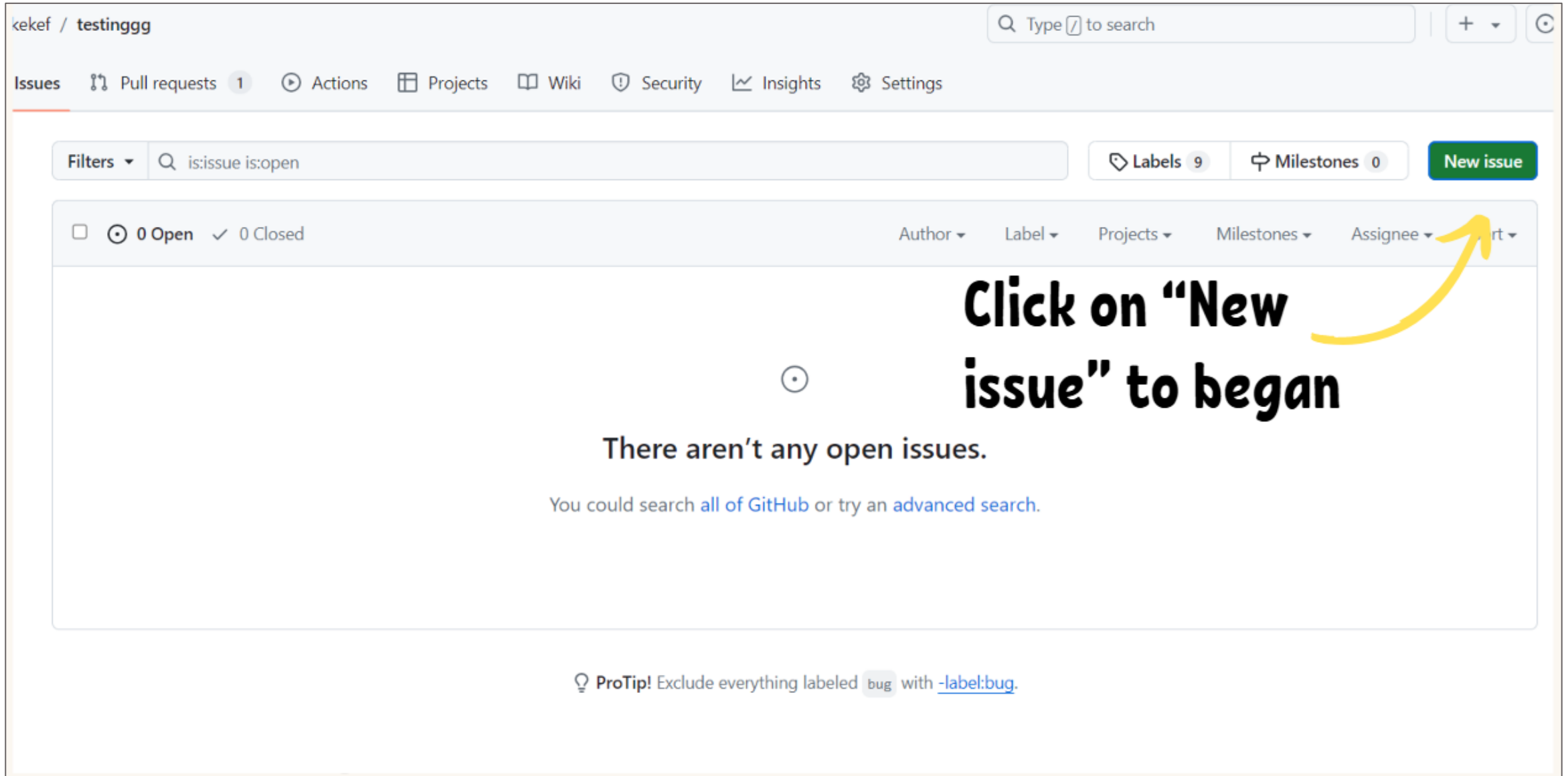
1. Creating and Managing Issues

- **Purpose:** Issues are used to track tasks, enhancements, and bugs in your project.
- **Steps:**
 - Navigate to Issues: Go to your repository and click the "Issues" tab.
 - New Issue: Click the "New issue" button.
 - Fill in Details:
 - Title: Provide a concise title for the issue.
 - Description: Describe the issue, including steps to reproduce, expected behavior, and screenshots if applicable.
- **Submit Issue:** Click "Create" to create the issue.

2. Using Labels, Milestones, and Assignees

- **Labels:**
 - Navigate to Issues: Go to the "Issues" tab.
 - Select Issue: Click on an issue to view it.
 - Add Labels: Click the "Labels" button and select or create labels to categorize the issue.
- **Milestones:**
 - Navigate to Issues: Go to the "Milestones" tab within Issues.
 - New Milestone: Click "New milestone", add a title and description, and set a due date if needed.
 - Add Issues to Milestone: Open an issue, click "Milestone", and select the milestone to associate with the issue.
- **Assignees:**
 - Open Issue: Open the issue you want to assign.
 - Assign: Click the "Assignees" button and select the team member to assign the issue to.

ISSUES: EXAMPLE



kekef / testinggg

Search Type to search

Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

Filters is:issue is:open Labels 9 Milestones 0 New issue

0 Open 0 Closed Author Label Projects Milestones Assignee Sort

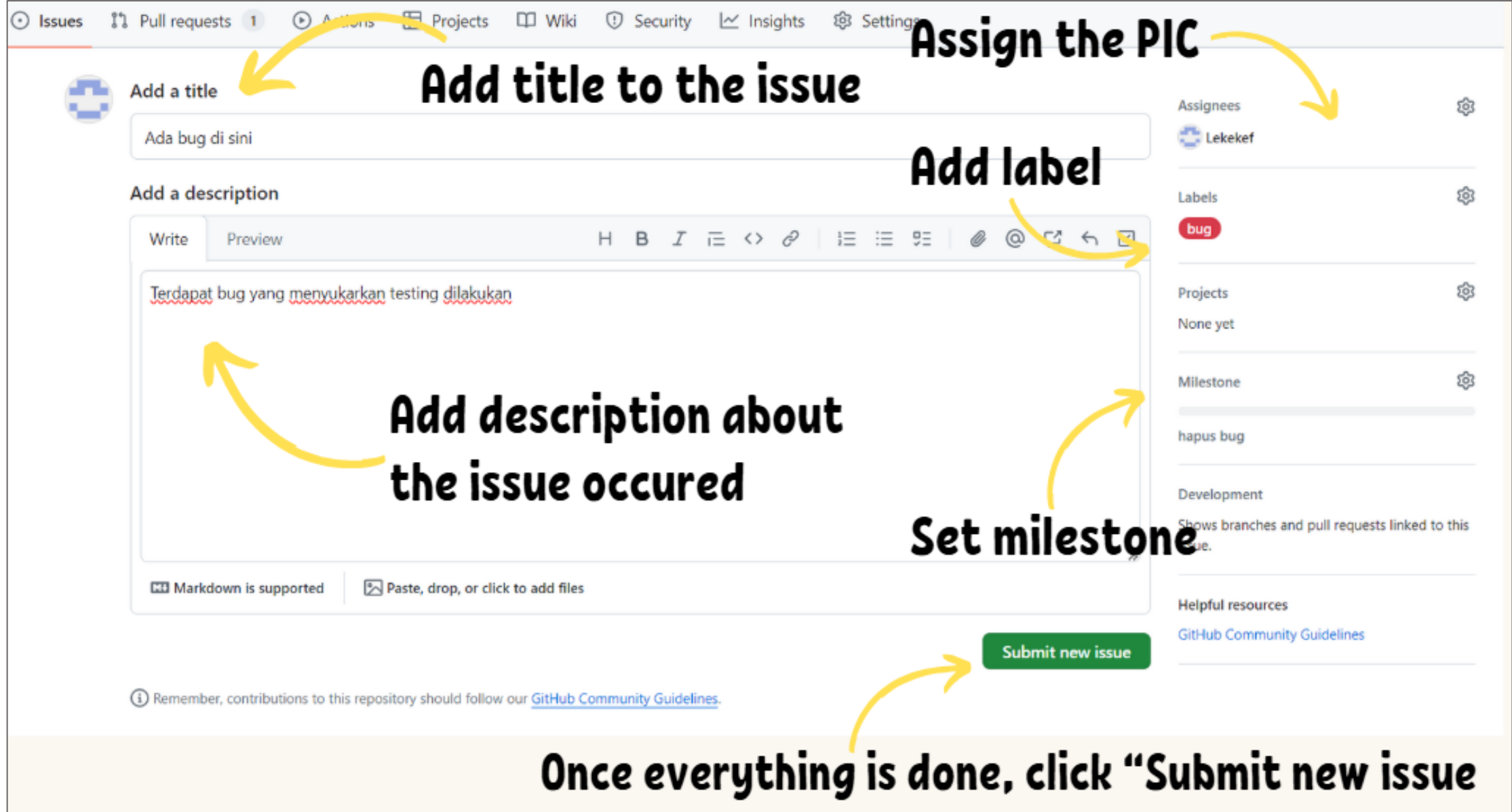
Click on “New issue” to began

There aren't any open issues.

You could search [all of GitHub](#) or try an [advanced search](#).

ProTip! Exclude everything labeled `bug` with `-label:bug`.

ISSUES: EXAMPLE



The screenshot shows the GitHub 'New Issue' form. The title field contains 'Ada bug di sini'. The description field contains 'Terdapat bug yang menyukarkan testing dilakukan'. The 'Labels' section has a 'bug' label selected. The 'Assignees' section has 'Lekekef' assigned. The 'Milestone' section has 'hapus bug' selected. The 'Submit new issue' button is at the bottom right. Annotations with yellow arrows point to various fields: 'Add title to the issue' points to the title field, 'Assign the PIC' points to the assignees section, 'Add label' points to the labels section, 'Add description about the issue occurred' points to the description field, 'Set milestone' points to the milestone section, and 'Once everything is done, click "Submit new issue"' points to the submit button.

Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

Add a title

Ada bug di sini

Add a description

Write Preview

H B I       

Terdapat bug yang menyukarkan testing dilakukan

Markdown is supported Paste, drop, or click to add files

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

Assignees

Lekekef

Labels

bug

Projects

None yet

Milestone

hapus bug

Development

Shows branches and pull requests linked to this issue.

Helpful resources

[GitHub Community Guidelines](#)

Submit new issue

Once everything is done, click "Submit new issue"

ISSUES: EXAMPLE

Ada bug di sini #2 **The issue has been successfully posted** [New issue](#)

[Open](#) Lekekef opened this issue now · 0 comments

 **Lekekef commented now** [Owner](#) [...](#)

Terdapat bug yang menyukarkan testing dilakukan

 **Lekekef added the [bug](#) label now**

 **Lekekef added this to the [hapus bug](#) milestone now**

 **Lekekef self-assigned this now**

People may add comments to help solve the issue

 **Add a comment**

Write Preview [H](#) [B](#) [I](#) [≡](#) [<>](#) [🔗](#) [📎](#) [📧](#) [🗑️](#) [↩️](#) [📄](#)

Add your comment here...

[Markdown is supported](#) [Paste, drop, or click to add files](#)

[Close issue](#) [Comment](#)

[Remember, contributions to this repository should follow our \[GitHub Community Guidelines\]\(#\).](#)

Assignees [⚙️](#)

 **Lekekef**

Labels [⚙️](#)

[bug](#)

Projects [⚙️](#)

None yet

Milestone [⚙️](#)

[hapus bug](#)

Development [⚙️](#)

[Create a branch for this issue or link a pull request.](#)

Notifications [Customize](#)

[🔔 Unsubscribe](#)

You're receiving notifications because you're watching this repository.

1 participant

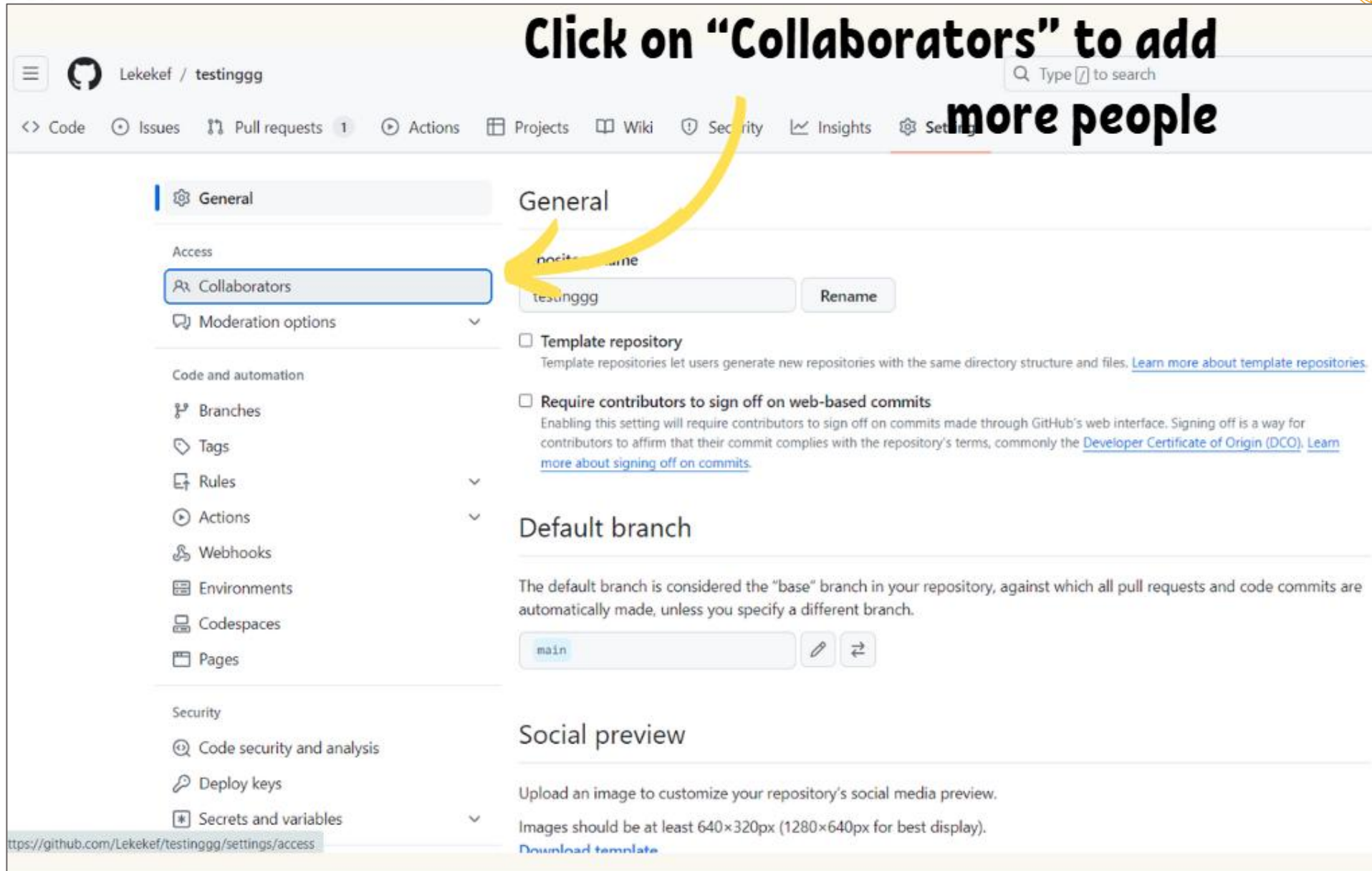
 [🔒 Lock conversation](#)

Managing Access Control and Permissions

- **Purpose:** Control who can view and contribute to your repository.
- **Steps:**
 - Navigate to Settings: Go to your repository and click the "Settings" tab.
 - Manage Access: Click "Manage access".
 - Add Collaborators: Click "Invite a collaborator" to add users and set their permissions (Read, Write, Admin).

SECURITY ON GitHub: EXAMPLE

Click on "Collaborators" to add more people



The screenshot shows the GitHub repository settings page for 'Lekekef / testinggg'. The left sidebar contains a list of settings categories: General, Access, Moderation options, Code and automation, Security, and Secrets and variables. Under the 'Access' category, the 'Collaborators' link is highlighted with a blue border. A yellow arrow points from the text 'Click on "Collaborators" to add more people' to this link. The main content area shows the 'General' settings tab, which includes options for 'Template repository', 'Require contributors to sign off on web-based commits', 'Default branch' (set to 'main'), and 'Social preview'.

Lekekef / testinggg

Code Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks

Environments

Codespaces

Pages

Security

Code security and analysis

Deploy keys

Secrets and variables

Template repository

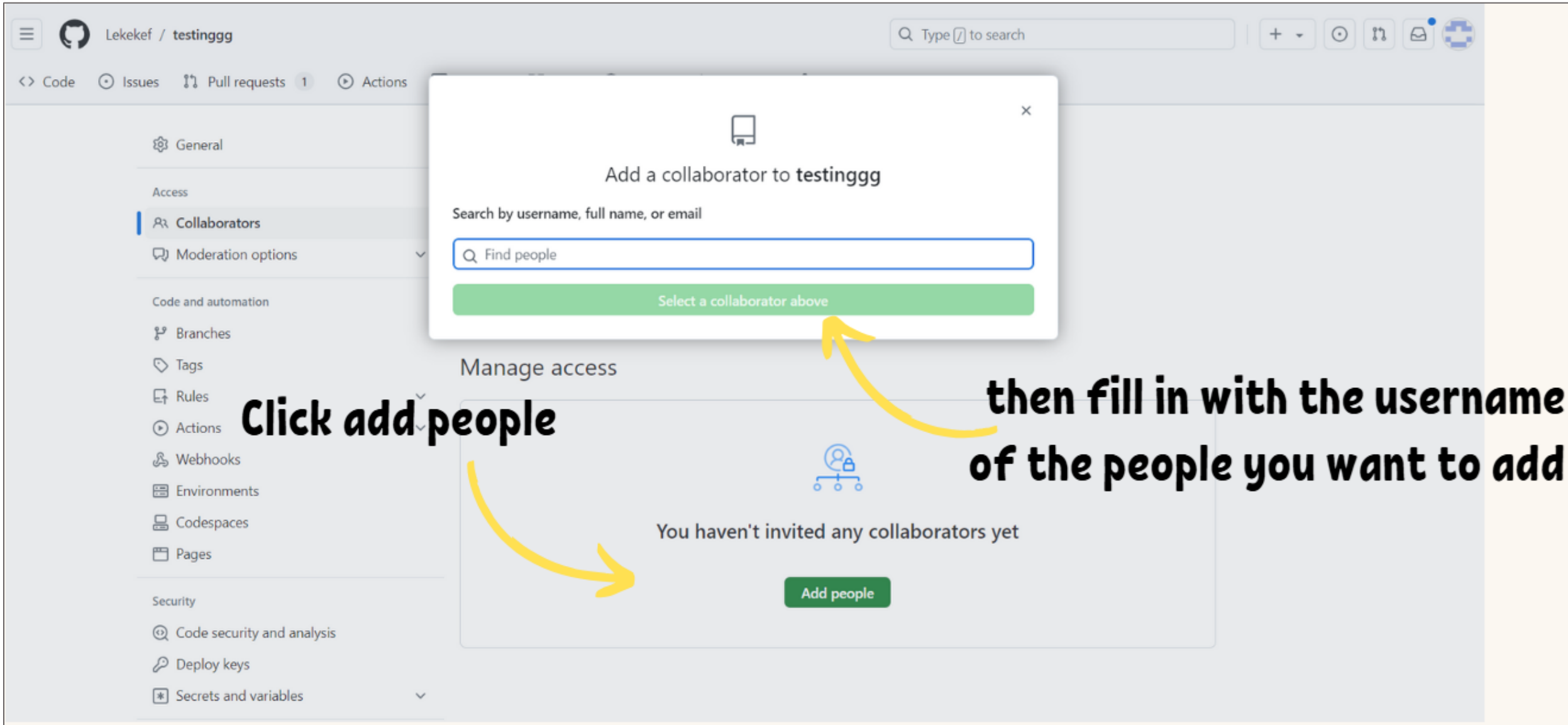
Require contributors to sign off on web-based commits

Default branch

Social preview

<https://github.com/Lekekef/testinggg/settings/access>

SECURITY ON GitHub: EXAMPLE



Lekekef / testinggg

Code Issues Pull requests 1 Actions

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks

Environments

Codespaces

Pages

Security

Code security and analysis

Deploy keys

Secrets and variables

Search Type / to search

+

×

Add a collaborator to testinggg

Search by username, full name, or email

Find people

Select a collaborator above

Manage access

You haven't invited any collaborators yet

Add people

Click add people

then fill in with the username of the people you want to add

SECURITY ON GitHub: EXAMPLE

Once done, we may view the list of people we want to add.

Manage access Add people

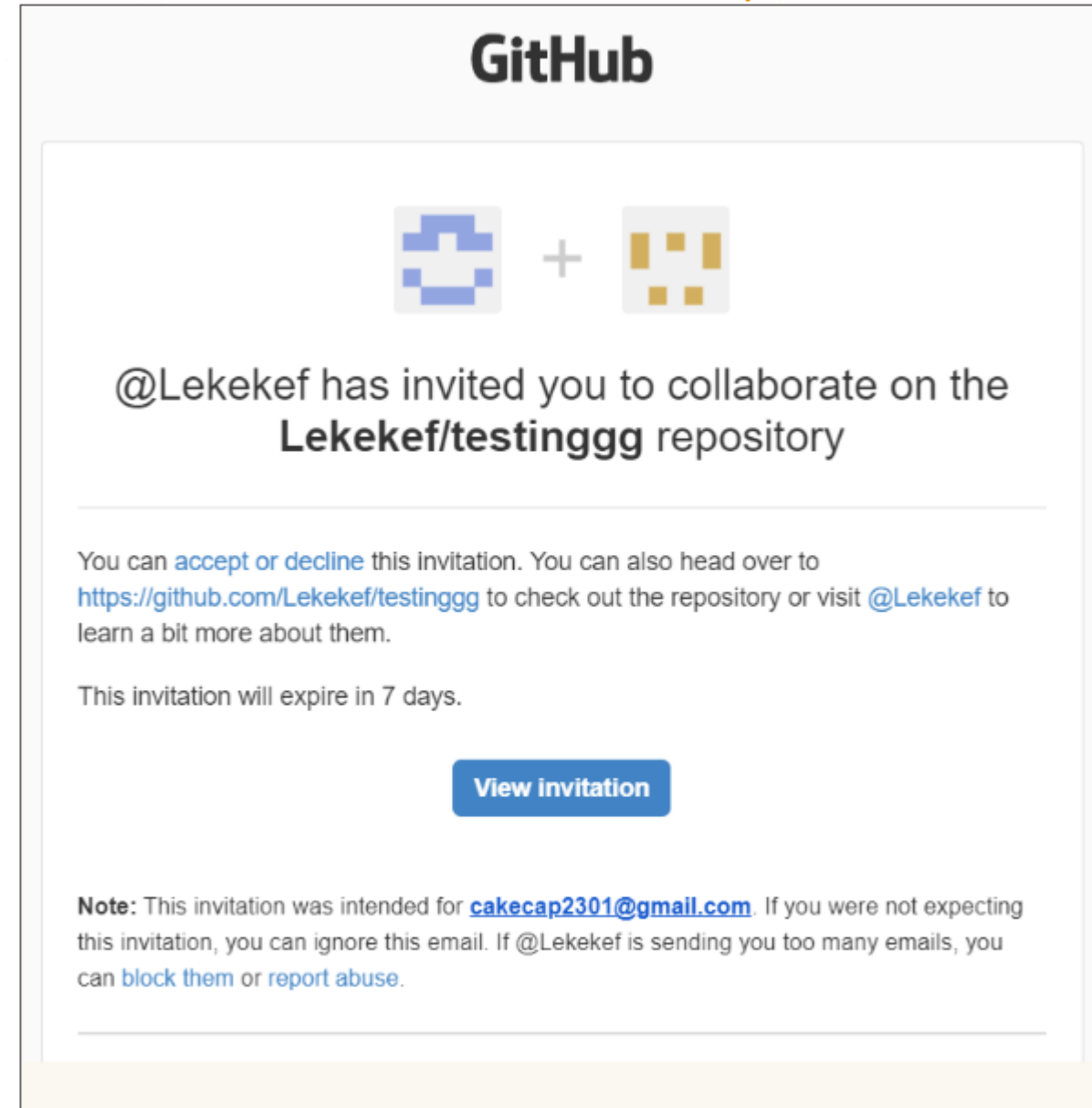
☐ Select all Type ▾

☐	 Cakecap Awaiting Cakecap's response	Pending Invite 	
--------------------------	---	--	---

But the invite is actually still pending, So, we need to wait until they accept it via email.

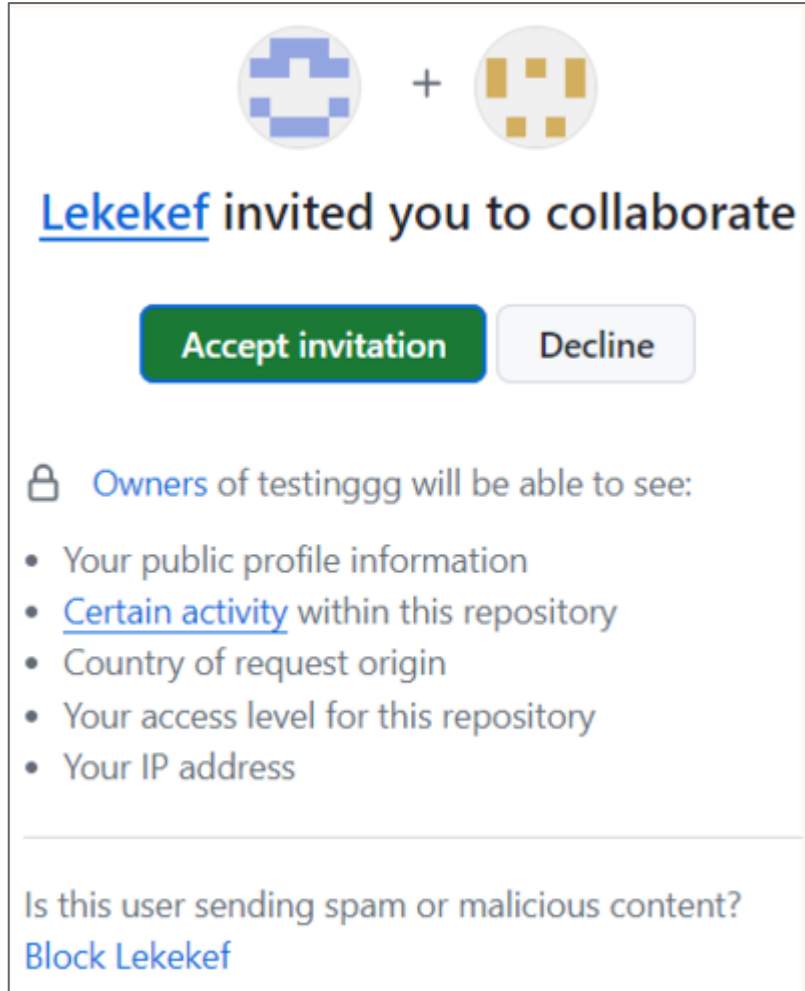
SECURITY ON GitHub: EXAMPLE

The invited people should get the invitation via email. Click on the “View invitation” to proceed.



SECURITY ON GitHub: EXAMPLE

Once the invitation is accepted, we may see a notification pop-up indicating the other user has been successfully added



The image shows a GitHub invitation interface. At the top, there are two avatars: a blue pixelated one and a yellow one, separated by a plus sign. Below them, the text reads "Lekekef invited you to collaborate". There are two buttons: a green "Accept invitation" button and a grey "Decline" button. Below the buttons, there is a section titled "Owners of testinggg will be able to see:" followed by a list of permissions: "Your public profile information", "Certain activity within this repository", "Country of request origin", "Your access level for this repository", and "Your IP address". At the bottom, there is a link that says "Is this user sending spam or malicious content? Block Lekekef".

 + 

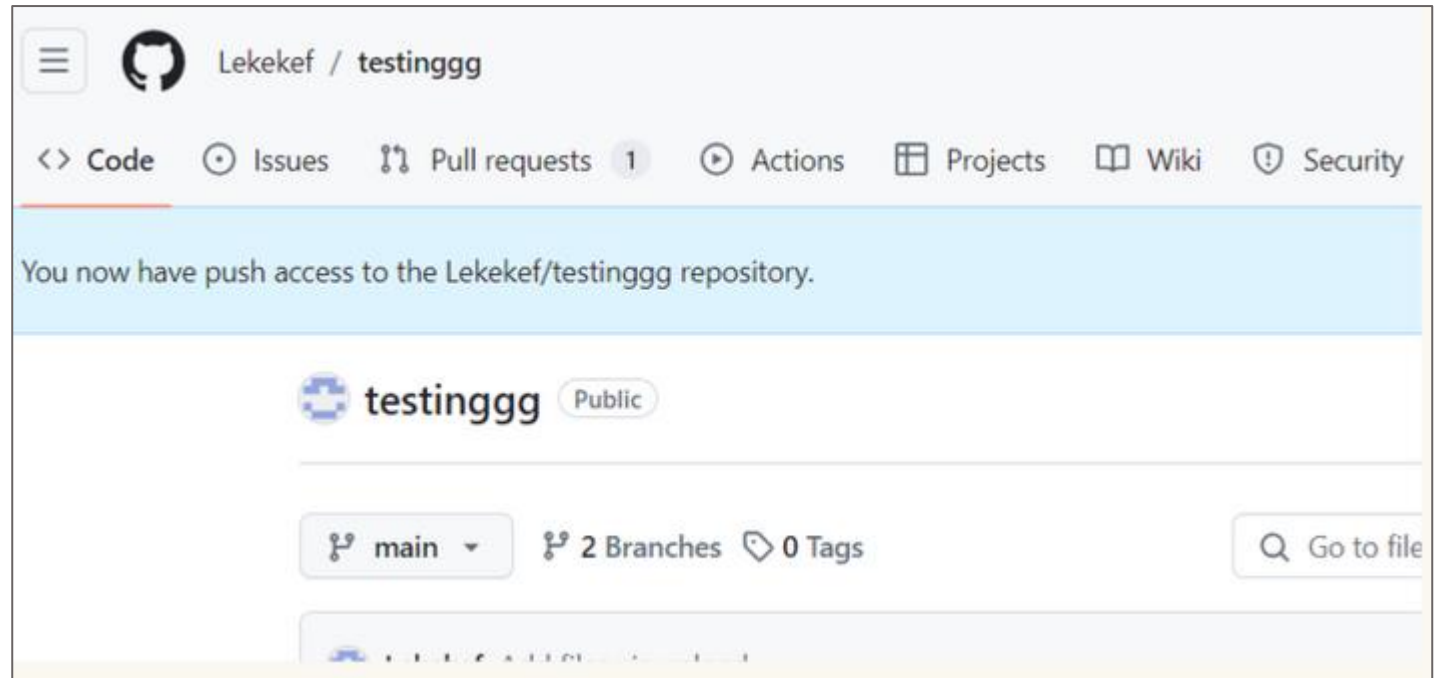
[Lekekef](#) invited you to collaborate

[Accept invitation](#) [Decline](#)

 Owners of testinggg will be able to see:

- Your public profile information
- [Certain activity](#) within this repository
- Country of request origin
- Your access level for this repository
- Your IP address

Is this user sending spam or malicious content?
[Block Lekekef](#)



The image shows a GitHub repository page for "Lekekef / testinggg". The page has a navigation bar with links for "Code", "Issues", "Pull requests" (with a count of 1), "Actions", "Projects", "Wiki", and "Security". Below the navigation bar, there is a blue notification banner that says "You now have push access to the Lekekef/testinggg repository." Below the banner, the repository name "testinggg" is displayed with a "Public" label. Below the repository name, there are buttons for "main" (with a dropdown arrow), "2 Branches", and "0 Tags". At the bottom right, there is a search bar with the text "Go to file".

  Lekekef / testinggg

[Code](#) [Issues](#) [Pull requests](#) 1 [Actions](#) [Projects](#) [Wiki](#) [Security](#)

You now have push access to the Lekekef/testinggg repository.

 testinggg [Public](#)

[main](#) [2 Branches](#) [0 Tags](#)

SETTING UP GitHub IN JETSON



Installing Git

- Install Git on your Jetson Orin:

```
sudo apt update
```

```
sudo apt install git
```

Configuring Git

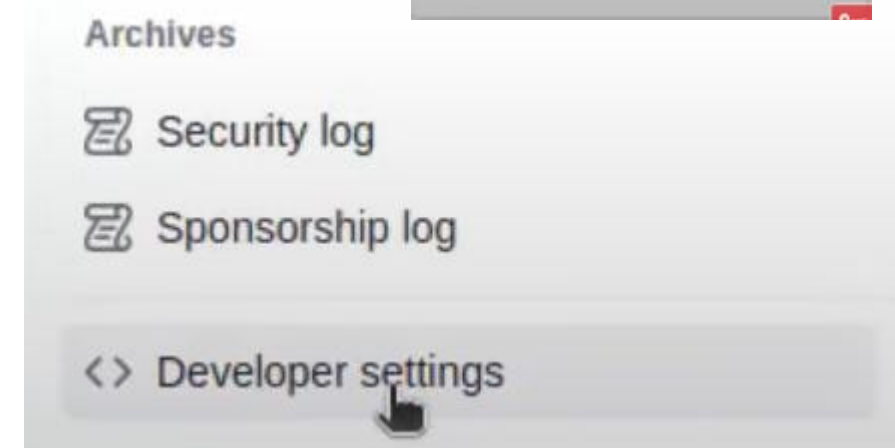
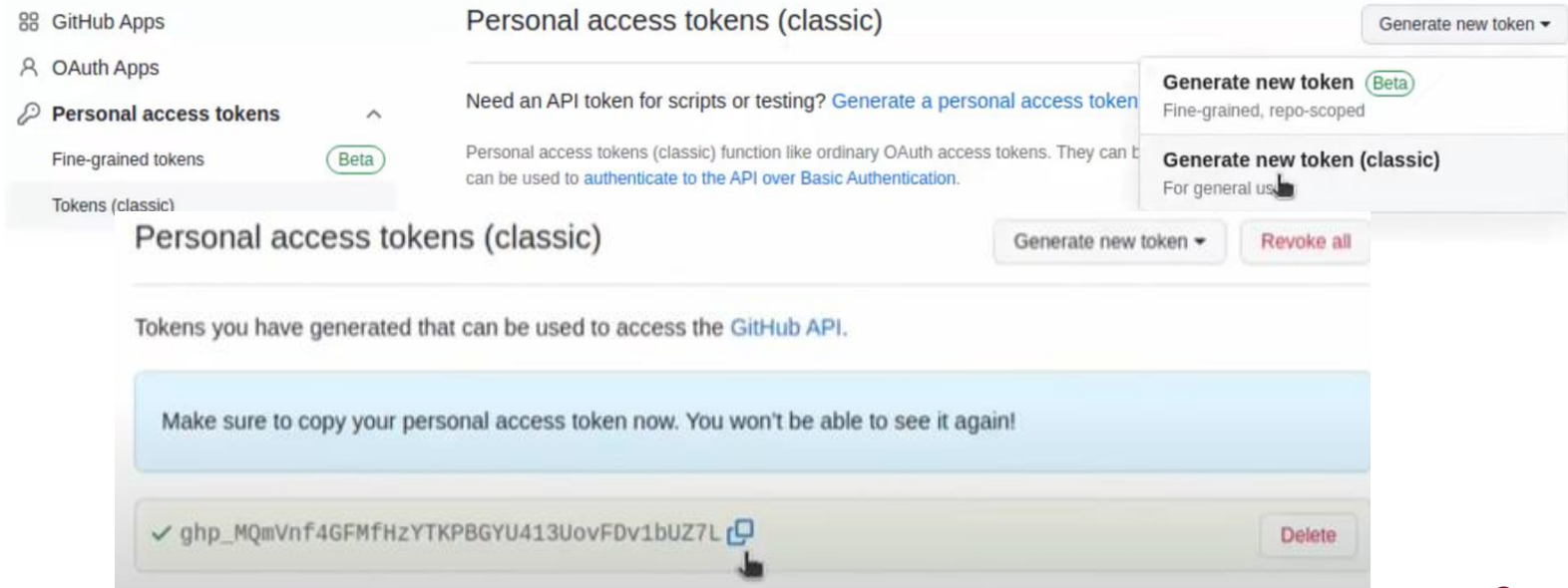
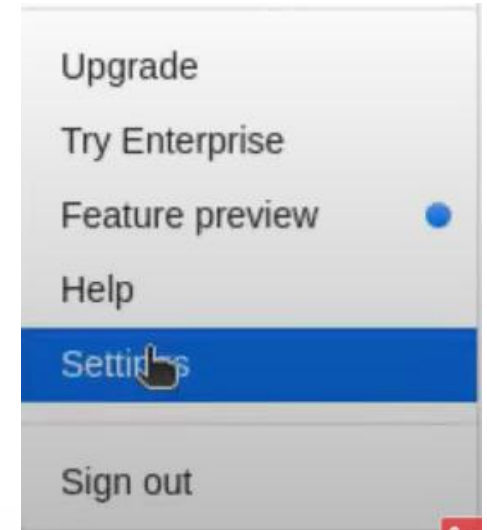
- Set up your Git username and email:

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your.email@example.com"
```


ACCESSING GIT

1. From a GIT client application, you authenticate to the GitHub server by using your **authenticate username** and a **Personal Access Token (PAT)**.
2. Select your profile icon in the top corner, select setting.
3. Then in the left panel select “Developer setting“
4. Click “Personal access tokens” , then generate new token (classic)
5. Fill in the note and select all the scope in generation and generate the token.
6. Copy the token (remember this only will show once)



JETSON TO AND FROM GitHub

- On the GitHub select your project, select Code on the right and then select the clone URL

- Use the 'git clone' command followed by the repository URL.

git clone <https://GitHub.com/username/repository.git>

- After the cloning process completes, you can navigate into the repository's directory:

Cd repository

- Adding Files to Git:

echo "print('Hello, Git!')" > hello_git.py

git add hello_git.py

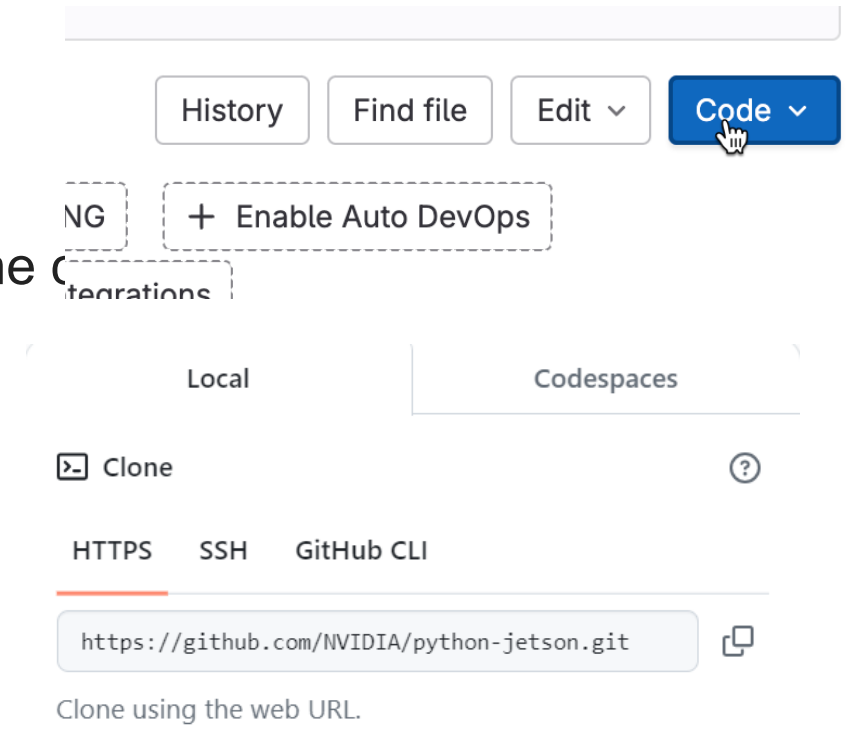
git commit -m "Initial commit"

- Pushing to a Remote Repository:

git status

git push origin main

- Use the token from last slide with your GitHub username.



```
codebind@codebind:~/workspace/hello_test$ git push origin main
Username for 'https://github.com': pknowledge
Password for 'https://pknowledge@github.com':
```

CONCLUSION



1. Professional Development

- Continue using GitHub to build a professional portfolio and engage in open-source projects.

2. Encouragement to Explore Further

- Apply what you've learned by working on personal or group projects.
- Seek Help and Collaborate: Don't hesitate to seek help from the GitHub community and collaborate with peers.
- Stay Curious: Always be curious and explore new features and best practices.

3. Empowerment Through Knowledge

- Understanding and effectively using GitHub will empower you to manage your projects better, collaborate efficiently, and continuously learn and grow as a developer.



THANK YOU



Centre for Artificial Intelligence and Robotics



research.utm.my/cairo/



cairo@utm.my / trainer's email